

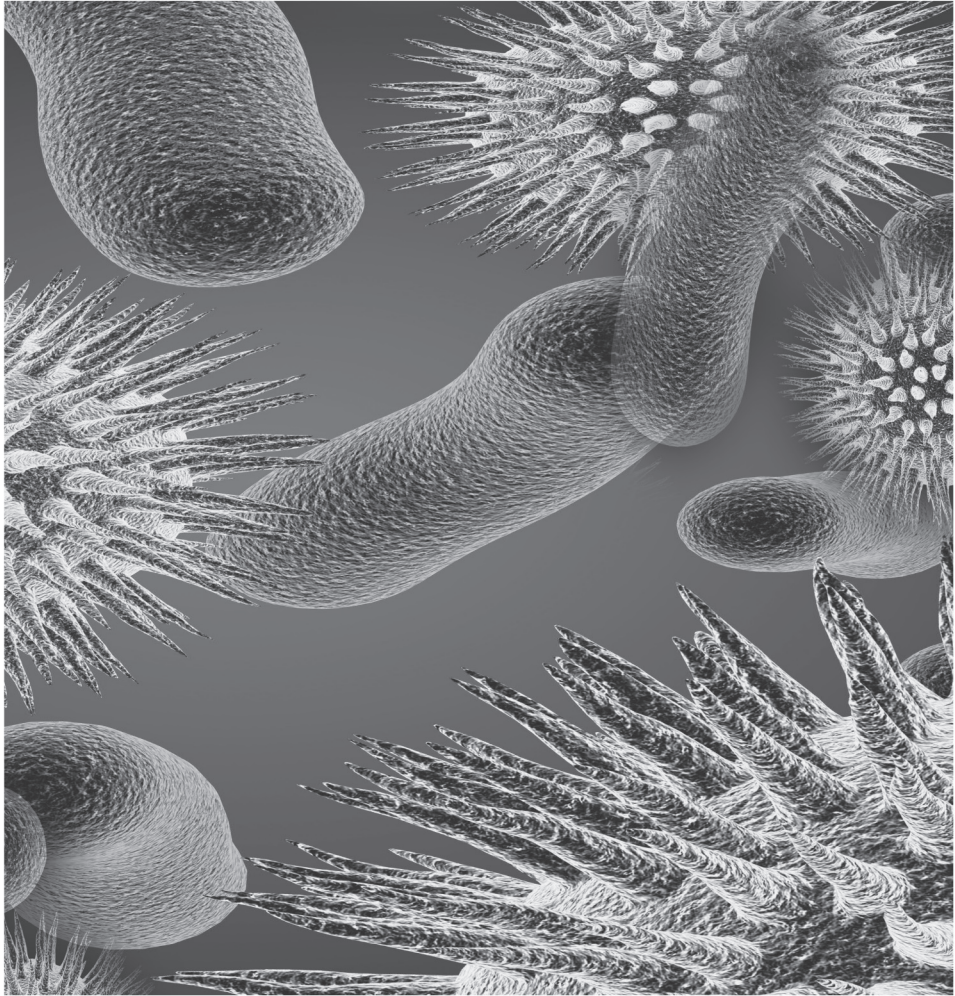
UNDERSTANDING
BIOLOGY
BIOTECHNOLOGY & DISEASE

Wendy St Germain, Peter Gribben



Quality ready-to-use
resources





UNDERSTANDING **BIOLOGY** BIOTECHNOLOGY & DISEASE

Wendy St Germain, Peter Gribben

This series was commissioned by Jennifer Gough, project management was by Piper Print Media, Gail McManus Graphics designed the cover, Final Page Editing and Design provided page design and typesetting services and David Dickson supplied the illustrations.

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Year 10–12 Biology teachers often report that their students lack basic knowledge of key scientific ideas, which hampers student progress in this subject. The authors of this series have developed activities for topics that, from their experience, they believe are poorly understood by students, as well as for concepts that are particularly crucial to an appreciation of the importance of modern scientific knowledge such as biomaterials, biotechnology and cloning.

In some units, students are required to investigate a topic and provide opinions, and these will make ideal homework or assignment work. Other units are fully self-contained and will test students' understanding of key scientific terms and how this language is used in context.

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Disease is described as disorder in the normal, healthy functioning of a living organism. It is a condition that reduces health and well-being in some way. Diseases can be categorised into several groups. Two main classifications are *acquired* and *congenital*.

Acquired diseases are those that people are not born with, but contract through other ways. While many of these are self-inflicted, such as smoking-related lung cancer and liver disease resulting from alcoholism, diseases can also be acquired through work and environmental conditions.

- 1 Name a different disease that could be acquired, and describe how it is acquired. _____

- 2 Why do people put themselves at risk of contracting diseases? _____

- 3 Animals are not the only organisms that contract 'illnesses'. Name a disease that affects the health of plants. _____

Congenital conditions are those that people are born with. These conditions may be diseases or syndromes. A *syndrome* is a group of symptoms that occurs together to produce a particular disorder (Greek *sundromē*, meaning 'to run together'). Many people think that a syndrome, such as Down Syndrome ('Trisomy 21', the result of extra genetic material from chromosome 21), is not a disease. One argument for this is that individuals with Down Syndrome often display cheerful dispositions. However, individuals who have Down Syndrome show varying degrees of mental retardation and roughly 40% have serious heart defects. (The damage is determined by how much extra chromosome 21 the person carries and which genes are involved.) Some people argue that it should be classified as a disease condition because of this impairment to health.

Some congenital conditions are apparent from birth, such as haemophilia (a blood disorder), Fragile X mental retardation (result of a damaged X chromosome), certain spinal disorders, muscular dystrophy and cystic fibrosis. (Cystic fibrosis is not always obvious at birth, onset may occur a few years later.) Other conditions are less obvious. For example, there is a form of bipolar disorder that results from an autosomal condition. (Autosomes are chromosomes that are not sex chromosomes.)

After identifying whether a disease is acquired or congenital, further groupings can be made. People often divide conditions into these eight groups:

- Congenital (inherited)
- Contagious (infectious)
- Degenerative
- Dietary
- Environmental
- Mental illness
- Occupational (industrial)
- Lifestyle

- 4 Classify the following conditions according to the eight groups above. (Note that some conditions fit into more than one category.)

- a** alcoholism _____
- b** Alzheimer's disease _____
- c** anorexia nervosa _____
- d** asbestosis _____
- e** asthma _____
- f** athlete's foot _____
- g** chicken pox _____
- h** diabetes _____
- i** foetal alcohol syndrome _____
- j** haemophilia _____
- k** heart disease _____



- l** HIV/AIDS_____
- m** influenza ('flu)_____
- n** leukaemia_____
- o** obesity_____
- p** Parkinson's disease_____
- q** phobia_____
- r** psychosis_____
- s** sexually transmitted diseases_____
- t** spinal disorders_____
- u** schizophrenia_____
- v** tooth decay_____

- 5** Comment on the importance of lifestyle in the prevention of disease. _____
- 6** Suggest why some conditions (in question 4) fit into more than one category. _____
- 7** Tooth decay and obesity may not be automatically classified as diseases. Why do you think they are included in the list in question 4? _____
- 8** The eight groups of illnesses are shown in the chart below. List a specific example and its symptoms for each group. Do not use examples already included on this worksheet.

	Type of condition	Example	Symptoms
D	Congenital		
I	Contagious		
S	Degenerative		
E	Dietary		
A	Environmental		
S	Lifestyle		
E	Mental illness		
S	Occupational		

- 9 Research** – For one of the conditions you listed in question 8, find out when it was first identified and how it is treated or prevented.



The word disease comes from the Latin words *dis* meaning ‘lack of’ and *aise* meaning ‘ease’ or ‘comfort’. Whenever a virus or other micro-organism impairs (harms, spoils) our health we can be in a state of disease.

- 1 Write each of these terms beside the correct definition below: *ailment, hypochondriac, pathogen, epidemiology, pandemic, contagious, epidemic*.
- a Widespread occurrence of disease in an area or country. _____
 - b An illness that can be spread by contact with others. _____
 - c A person who is overly concerned about their health, always thinks they are ill. _____
 - d An illness or medical complaint. _____
 - e Large scale spread of a given disease across the world (global). _____
 - f Study of the spread of disease. _____
 - g A disease-causing microbe. _____

There are many causes of disease, including genetics, parasites, viruses, bacteria, sexual transmission and obesity.

Genetic diseases are inherited from one or both parents (who are called carriers). These diseases are ‘programmed’ to occur through information coded in our DNA (our genetic material). Most genetic diseases are recessive, meaning they require a ‘dose’ of the recessive gene from each parent. The fact that both parents need to donate a recessive gene makes many genetic diseases much rarer. Sick-cell anaemia is an example of a recessive, blood-related condition that can be inherited. This disease causes the red blood cells to collapse into a sickle (crescent) shape, which reduces the amount of oxygen that can be carried through the body. The cells also tend to clog up the blood vascular system which can be dangerous. Genetic diseases are not contagious (not infectious, cannot spread to other people).

- 2 Why would the fact that both parents need to provide a gene make a condition rarer?

- 3 **Research** – Find out the name of another genetically inherited disease condition, and its effects.

Parasitic diseases are spread by parasites such as worms. Tapeworms and liver flukes are examples of parasitic organisms that cause disease. (A typical tapeworm is shown in figure 1.) After infection, they reduce the health of the individual by absorbing nutrients that the host body needs to survive. Tapeworms are difficult to get rid of because they grow in sections called proglottids. Proglottids break off and each one can grow into a new tapeworm, each of which can produce thousands and thousands of new tapeworms. (One estimate suggests that about 130 000 new worms can grow each year from a single worm!).

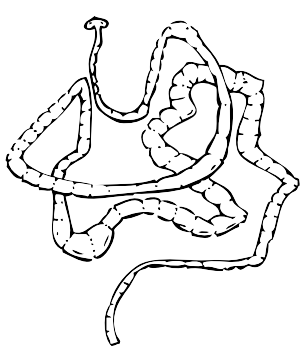


Figure 1

- 4 Tapeworms are hard to destroy once an individual is infected. If, under ideal conditions, a tapeworm could produce 130 000 proglottids every year, how many could it potentially produce over 25 years?

- 5 How might a tapeworm affect an infected human? _____
- 6 *An organism taking nourishment from another (host) organism, damaging the host in the process* – what does this describe?

- 7 **Research** – Find out about a different parasite that can affect humans. How are humans infected? How does the parasite affect its host?



Viral diseases – The word *virus* is derived from the Latin word meaning ‘poison’. Viral diseases include influenza, measles and smallpox, and are often contagious. Viruses are tiny ‘packages’ and are usually considered ‘not living’. They are little more than mobile packets of genetic material (DNA or RNA) that require a host (living organism) before they can be fully activated. Viruses lack the cell components, such as cytoplasm and its contents, that are necessary to reproduce their genetic material. Viruses have no ‘assembly materials’, just the genetic material that directs their replication after taking over a healthy cell. Figure 2 shows influenza viruses.

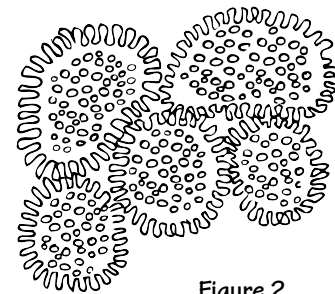


Figure 2

- 8 Research** – Find an example of a different viral disease and describe its effects.

Bacterial diseases are caused by living microbes. Figure 3 is a simplified diagram of a common bacterium. Bacteria are often contagious through being airborne or by ‘waiting’ for the right conditions.

Tetanus is a bacterial condition that ‘waits’ for the right conditions. Many farm animals have contracted ‘lock jaw’ from stepping on a nail in an area where tetanus spores are present. The microbes need a puncture wound to enter the body. Fortunately, veterinarians can treat animals who suffer a puncture wound with a tetanus shot. This simple injection has saved many lives. Many people ensure their tetanus immunisations are up to date since humans can also contract this disease.

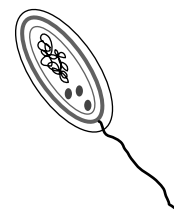


Figure 3

Bacteria can travel through water, as well as air. Cholera is a waterborne disease that kills many people. It is often found in places where disaster strikes, such as flooding, where many bodies may rot in flood waters. The ‘flu is another example of an airborne disease. The damp, sneeze droplets of affected people are an excellent way of transmitting airborne disease.

- 9** Suggest why tetanus infection is often called ‘lock jaw’.
- 10** Identify two media through which bacteria can travel.
- 11** Can tetanus be fatal? Explain your answer.

Venereal disease is an example of a sexually transmitted disease (STD). A person must come in direct contact with the carrier, through sex, in order to be infected. (*Venereal* comes from *Venus*, the Roman goddess of love.) Many people are too embarrassed to admit to their doctors that they have a venereal disease, and this can help the disease spread. Many venereal diseases can be treated, but only if the carrier is aware of the disease, and willing to get medical help. One of the most common venereal diseases is caused by a bacterium that results in gonorrhea.

- 12 Research** – Find out more about gonorrhea or another STD.

- 13 Research** – Many sexually transmitted diseases are curable. Find out about an STD that is incurable.

Obesity, being overweight and insufficient exercise lead to many deaths every year. Obesity causes a number of disease conditions that could be prevented by a healthy lifestyle, such as diabetes and heart disease. These two examples can also be inherited, but for many people, they are brought on by lifestyle choices, and therefore are preventable.

- 14** What is the difference between being overweight and obese?



Fred Hollows (1929–1993): A man with a vision

Professor Fred Hollows is known and admired by many people for the incredible number of lives he improved by reducing blindness. Fewer people are aware that this New Zealand-born ophthalmologist (eye specialist) was the Head of a Sydney hospital's Eye Surgery Department and a pioneer in the treatment of *trachoma*.

Fred Hollows was appalled by the number of Australian Aborigines suffering unnecessary blindness as a result of poor health and unclean living conditions. He set to work enlisting aid to save their sight. By the 1970s, he had reduced blindness in many Aboriginal communities by up to 50%. The cause of this blindness was *Chlamydia trachomatis*, a micro-organism with both viral and bacterial characteristics. *C. trachomatis* causes the condition *trachoma*, the main cause of human blindness around the world. The condition is extremely contagious, as the micro-organism is found in the runny 'weeping' discharge it creates as the eye becomes increasingly inflamed. The discharge is easily transferred to fingers, tissues and other surfaces. It also travels easily via flies as they move from person to person.

This is an ancient condition, dating as far back as the Egyptian Pharaohs. It causes scarring on the fine membrane coating the inner eyelid and eye surface (conjunctiva). It is extremely painful and the inflammation and scarring continues until finally, if left unchecked, blindness results. *Trachoma* continues to be a problem in poor countries where sanitation, access to clean water and good hygiene are inadequate.

- 1 Suggest ways in which improved sanitation plays a role in eliminating *C. trachomatis* and other hygiene-related conditions.
 - 2 Suggest why people working with human remains often wear gloves and masks.
-
- 3 In your workbook, draw and label a diagram to show how trachoma can be transmitted in a desert community. Include sources of contamination and recontamination.
 - 4 **Research** – Find out more about the work of Fred Hollows and the foundation that bears his name.

Avian Influenza – Bird Flu

Avian bird 'flu has decimated bird populations across Asia (*avis*, meaning 'bird'). It could reach Australia and other countries via migratory birds. The virus, an avian version of influenza, is extremely virulent (meaning 'severe, rapid and malignant') and has already killed many thousands of birds and many suspected carriers have been destroyed. Slight mutations can cause the disease to cross species. Some scientists fear this will happen or has already happened. By mid-2005, over 50 people in south-east Asia had died of suspected bird 'flu. The disease was first identified in Italy in 1898, but has since spread across the world, with the latest outbreak originating in Asia.

- 5 Avian influenza has also been known to affect pigs. Suggest why the pigs might be affected, rather than other farm animals. Is it possible that other farm animals could be affected?
-

Bird 'flu is transmitted through the air (airborne, inhaled) or by coming into contact with contaminated droppings. There is also a mild form of avian influenza, but the form most publicised is the highly virulent form, which can reproduce so quickly that it can kill a bird within hours of contamination. The World Health Organisation (WHO) has reported that a single gram of the virus can kill over a million birds. Influenza is an RNA virus, which means it has RNA as its genetic material or that it passes through an RNA stage during replication. (RNA is a genetic material which contains uracil as its nitrogen base instead of thymine, which is found in DNA.) There are 15 subtypes of influenza virus, but three main groups which are categorised as A, B and C. The strains are identified by protein 'coats' on the outside of the molecule. The immune system uses these outer coats to identify invading organisms, since the body has its own, recognisable proteins (antigens) and all others are considered invaders. The coats are 'read' like a Braille system and if the bumps are not identified as 'self', the body sets about destroying the molecule with the non-self outer coat.

- 6 Outline how bird 'flu can be transmitted. _____
 - 7 Use the previous information to explain why drugs must be used to suppress the immune systems of people who have received organ transplants.
-



Influenza A and B affect humans (and other animals) much more aggressively than group C does. The 'super 'flu' we often hear about is a strain of influenza. Influenza A has two subtypes, of which one has an extremely high mortality rate among humans. Influenza C can be present in a person without any ill effects presenting themselves.

8 What is meant by *mortality rate*?_____

Mutation means to change or alter from an original form. It occurs in two ways: by *antigenic drift* and *antigenic shift*. As the names suggest, drift is a more gradual change, 'a bit here, a bit there', until it has come to resemble the genetic material of a species. Shift is more dramatic – the genetic material makes a full shift, not in steps, from associating with one species to the next. Some genetic materials have very low mutation rates while others have very high rates. Unfortunately, RNA viruses have high mutation rates, which means they can 'jump' species boundaries much faster and more easily due to their ability to change just enough to suit another species. The influenza strain that the public worries about is called H5N1. (The letters and numbers indicate their chemical components.) The great concern about the highly virulent H5N1 strain lies in the fact that it easily 'borrows' genetic material from other species. This ability to incorporate a species' genes into its make-up is what allows it to mutate and jump species (the more genetically similar to a species it becomes, the more able to affect that species it is).

9 What is meant by *virulent*?_____

10 It has been suggested that during the 1990s, Chinese farmers used human antiviral influenza vaccines on poultry (with government approval). If this is the case, there is a great risk that if people contract bird 'flu, they will be unable to benefit from a vaccine. Suggest why this is the case.

11 A sudden, permanent change in hereditary material – what does this describe?_____

12 Suggest why domestic poultry are at risk if there is just one infected bird present._____

13 Excluding the direct medical effects of killing birds, discuss the other effects bird 'flu, or any other disease outbreak, has on society.

14 Influenza affects the whole body but is called a *respiratory illness*. What does this mean?_____

15 Describe the symptoms of influenza._____

16 **Research** – Severe Acute Respiratory Syndrome (SARS) is another RNA virus that has been of great concern to the world's health organisations. Outline what type of virus causes SARS, and what the symptoms of this disease are.

17 Mark the following statements as true or false. Then correct the false statements so that they are true.

a Genetic diseases are contagious._____

f Influenza C is highly virulent._____

b Influenza is a viral disease._____

g All sexually transmitted diseases are curable if you seek treatment._____

c Bird 'flu is caused by a parasite._____

h Viruses need a living host in which to reproduce._____

d Trachoma is an infection that can lead to blindness._____

i The outer coats of viruses are made of protein._____

e Trachoma has both bacterial and viral characteristics._____

18 **Research** – In 2007, the Australian racing industry suffered great financial losses when hundreds of horses contracted equine influenza. Write a brief report about this influenza strain and its effects on the financial aspects of racing that year.



- 1** Many scientific words have Latin or Greek origins. The word bank below contains 12 words you will encounter when learning about disease. Write each word beside its Latin (*L.*) or Greek (*Gk*) origin and meaning to help you understand how the term developed.

Word bank: *acute, chronic, colonise, contagious, disease, infect/infection, local, primary, secondary, sign, symptom, systemic*

- a** *Gk sumptoma* – a thing that has befallen someone – _____
- b** *Gk chronos* – of or over time – _____
- c** *L. dis + aise* – lack of ease or comfort – _____
- d** *L. contagio* – contact – _____
- e** *L. signum* mark, represent, symbol _____
- f** *L. inficere* – to stain or taint – _____
- g** *Gk synistani* – placed together, throughout – _____
- h** *L. primarius* – from *primus*, meaning first – _____
- i** *L. colonia* – settlement, to farm, cultivate – _____
- j** *L. secundus* – to follow – _____
- k** *L. localis* – from *locus*, meaning place – _____
- l** *L. acutus* – sharp, like a needle – _____

- 2** Now use the word bank from the previous activity to complete the passage below.

_____ is defined as a change from normal, good health to a state of reduced or deteriorating health. Often, we can detect that our health has been affected by the presence of signs and symptoms. A _____ is a change that the affected person can feel; for example, aching muscles. The person notices a decrease in wellness, but it cannot be tested or measured using scientific means. A _____ is a measurable change in a person's health. This includes things like detecting an increase in the number of white blood cells circulating in the bloodstream (white blood cell numbers increase at times of _____). Illness that is inherited (genetic) is called non-communicable. This means that it will not spread through exposure to an affected individual. Non-communicable diseases are also called non-transmissible, meaning they are not transmitted by exposure to the sick person. Communicable diseases are also called infectious or _____. Illness occurs when the body is _____ by disease-causing micro-organisms. Certain terms are used to give more detail about infection. When infection is restricted to a small area of the body, it is called a _____ infection. Infection that spreads throughout the whole body is called _____ infection.

_____ infection always leads directly to an illness. Chicken pox is an example of this kind of infection. Sometimes we develop an infection as a result of having caught something else first. This is called a _____ infection. Such a condition is a type of 'after effect' of having survived the first. Shingles is an example of a health problem that can develop after someone has caught chicken pox. The words *acute* and *chronic* are also used when describing infection. _____ infections are caught, develop and leave in a relatively short space of time (compared to the lifespan of a person). The patient catches the infection, suffers for a short time then recovers. Lingering ailments can recur or remain for very long periods of time. They develop gradually and recur or last for a long time. These are called _____ conditions.

- 3** Mark the following statements as true or false. Then correct the false statements so that they are true.

- a** Communicable diseases can spread through a community. _____
- b** Chronic conditions tend to linger over long periods. _____
- c** The type of disease contracted depends on the organism involved. _____
- d** A symptom is a measurable example of a person's illness. _____



- e** Some diseases are exclusively plant-related or animal-related._____
- f** HIV/AIDS is an example of a non-communicable disease._____
- g** The fungus Athlete's foot is an example of a systemic condition._____
- h** Disease can spread through a replicating (reproducing) organism, virus or even a piece of genetic material that codes for it._____
- i** Tuberculosis (TB) is an example of a very old communicable disease that still kills many millions of people every year._____
- j** A disease is always caught from someone or something already carrying it._____
- k** In optimum conditions, one micro-organism can colonise an entire organism._____

4 'The incidence and type of diseases vary, depending on geographical area' – discuss.

5 'Two people of the same age with the same disease, living in different countries, might have totally different outcomes to the disease' – discuss.

6 Some diseases are restricted exclusively to a specific animal species. Some diseases can be contracted by different animal species, through interaction with infected animals. Find one example of a disease that can be contracted by humans, through interaction with infected animals of a different species. Briefly describe whether your example is viral or bacterial and its effects on both species involved.



Diphtheria is a highly contagious disease that causes inflammation of the heart and nervous system. Prior to the advent of a specific antibiotic to treat this disease, the mortality rate for people who contracted the illness was about 35%. Antibiotics are only useful against bacteria, they do not affect viruses.

- 1 What is meant by ‘a 35% mortality rate’?
-
-
-
- 2 Define the following terms.
- a Contagious
- b Specific
- c Prior

The column chart (figure 1) shows the number of deaths due to this disease over a period of time in one particular country.

- 3 Write a brief analysis of the trends shown in the graph.
-
-
-
-
-
- 4 When do you think the antibiotic was first introduced? Explain your answer.
-
-
-
-

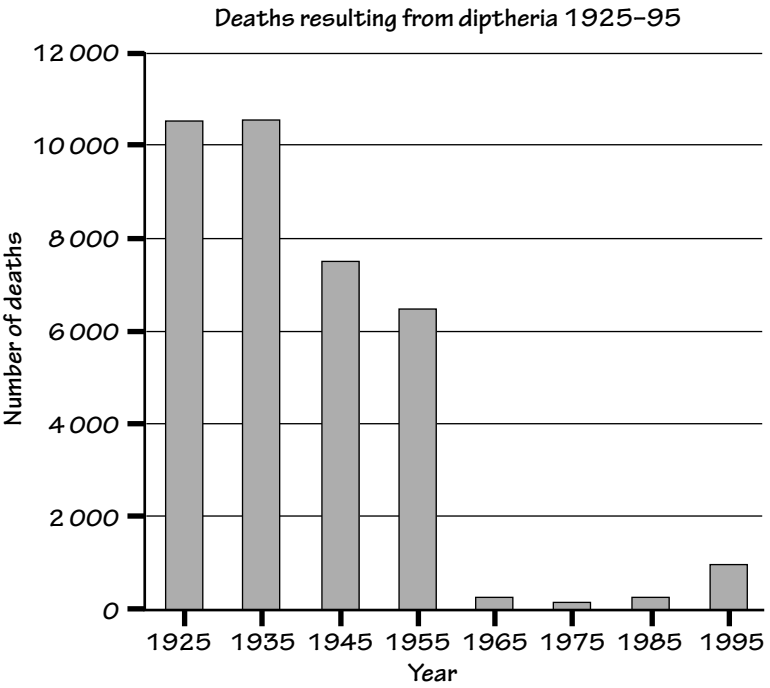


Figure 1

- 5 When did the use of the antibiotic first *control* the disease? Explain your answer.
-
-
-
-
-
- 6 Was this disease caused by a virus or a type of bacteria? Explain your answer.
-
-
-
-
-
- 7 In recent years, scientists have been experimenting with new treatments for diphtheria. Suggest reasons for this.
-
-
-
-
-
- 8 What do you think the number of deathswill be in the year 2010? 2020? Explain your answers.
-
-
-
-
-
- 9 List a symptom of diphtheria.
-
- 10 Name one antibiotic in current use.
-



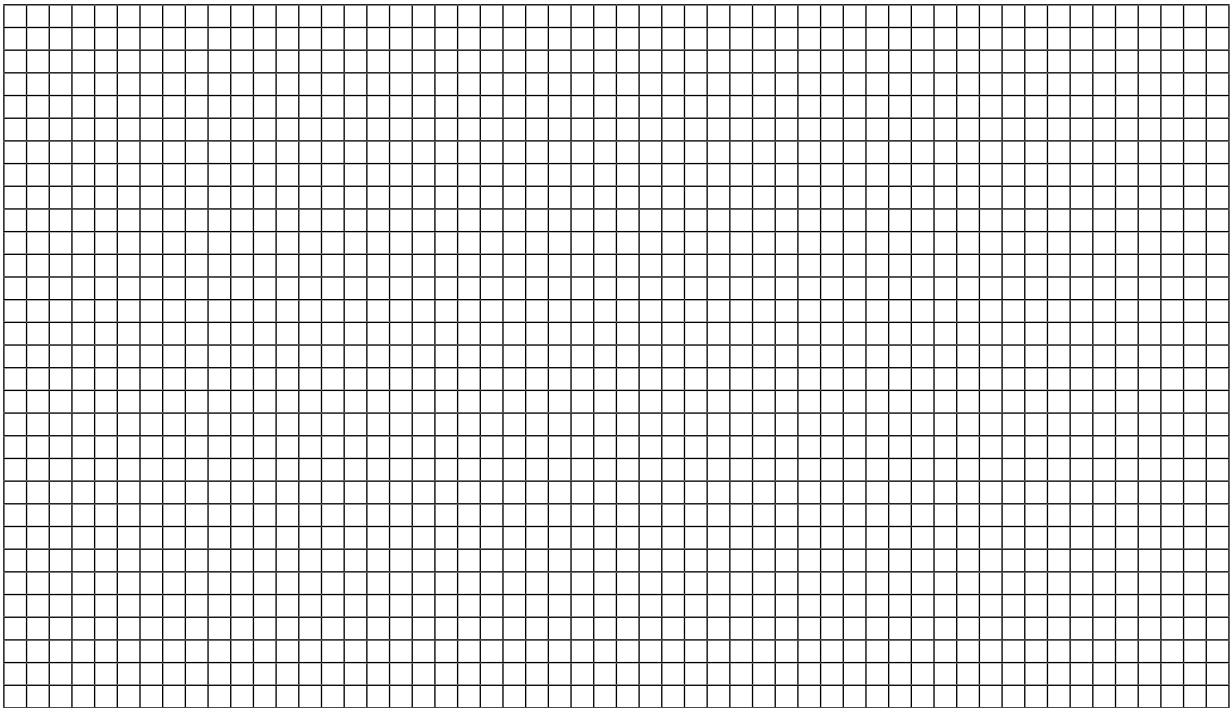
A medical researcher deliberately infected himself with bacteria known to cause a specific disease. His blood was analysed before the infection and then every day afterwards, and the number of bacteria was analysed. Any display of symptoms of the disease was also noted. The results are shown below in table 1. An asterisk (*) means that there were symptoms of the disease present on that day.

Time (Days)	0	1	2	3	4	5*	6*	7*	8*	9*
No. of bacteria	0	100	200	400	600	900	2600	5900	7800	7900

Time (Days)	10*	11*	12*	13*	14*	15	16	17	18	19
No. of bacteria	6100	3900	2200	1200	700	600	400	200	100	50

Table 1

- 1
- Draw a ‘best fit’ line graph using the data supplied in the table. Label the axes and write a title for the graph.



- 2
- What is meant by the term *symptoms*?_____
- 3
- On the graph, show when the symptoms were apparent.

The incubation period is the time taken from infection to the first symptoms of the disease.

- 4
- Show the incubation period on the graph.
- 5
- What is the smallest number of bacteria that caused symptoms in this subject? Explain your answer.

- 6
- Suggest reasons why the researcher chose to infect himself. What concerns would you have about this method of experimentation?



Invasive procedures

In medicine, a procedure is a process carried out on a patient that is designed to lead to an improvement in the patient's health. The procedure can be designed to either investigate or fix a problem. By their very nature, major operations are invasive procedures. Any surgical procedure that involves 'invading' the body in a way that can lead to severe trauma and shock is considered invasive. Examples include: traditional open-heart surgery, which requires a large incision to be made in the patient's chest, followed by breaking open the breastbone; hysterectomies (the complete removal of a woman's uterus); and hip replacements.

- 1 Define *trauma* and *shock*, as used above. _____
- 2 List a different example of an invasive procedure. _____

Wherever possible, invasive surgery is avoided or modified in order to reduce stress and trauma. For example, women who require operations involving the uterus often choose laparoscopy (keyhole surgery) as an alternative to a full hysterectomy. Sometimes a full hysterectomy is necessary, for example, to treat cervical cancer and endometriosis.

- 3 Women who undergo full hysterectomies require hormone replacement therapy (HRT). What does this tell you about the uterus? _____

Non-invasive and minimally invasive procedures

These either do not involve physically entering a patient's body or minimise the entry as much as possible. The advantages of non-invasive and minimally invasive surgical procedures are great to the patients, the surgeon and the tax-paying public. Considerable amounts of money are spent annually on our health system, of which large amounts go toward post-operative care (*post*, meaning 'after'). The longer a person remains in hospital, the greater the cost and the fewer the beds available for other patients.

- 4 Discuss the financial and emotional costs involved in hospitalising people. _____
- 5 List three examples of procedures that are now minimally invasive, but were invasive in the past. _____

For the patient, the less time spent under anaesthetic and in surgery, the quicker the rate of recovery.

- 6 Outline at least two reasons why anaesthetic is often used during surgery. _____

Non-invasive procedures reduce risks, both emotionally and physically, by placing the body under far less stress than invasive procedures. Consider exploring the possibility of a brain tumour. Invasive investigation requires opening the braincase and handling parts of the brain, which places the exposed tissue at risk of damage and contamination. Non-invasive techniques use scanning X-rays or other painless devices, at less risk and cost than opening the braincase. These technologies produce a picture of the patient's insides without having to physically cut into the tissue. X-rays are not without their risks – prolonged exposure can cause tissue damage and cancers.



Figure 1

- 7 Describe how the development of these technologies improves the diagnosis and treatment of brain tumours. _____

Ultrasound – 'Seeing with sound'

Ultrasound uses sound waves, higher than those detectable by the human ear, to produce an internal image (or *sonogram*). A water-based conducting gel is spread over the outer area of the tissue being investigated. A hand-held probe, called a *transducer*, moves smoothly across the gel, directing the path of the waves around a localised area. As the high intensity waves bounce off tissues, an image is created by the console to which the transducer is attached.



Ultrasounds do not produce clear images, so technicians are specially trained to read them. The lack of clarity is compensated for by the fact that the procedure does not emit ionising radiation (unlike X-rays). This means that there are currently no known side effects or risks associated with repeated use of ultrasound.



Figure 2

8 Research – Define *ionising radiation*.

9 Research – List two examples of ionising radiation.

10 Suggest why ionising radiation can be dangerous.

11 Pregnant women often have sonograms taken of their developing babies. Why?

Therapeutic uses – Ultrasound and healing

Ultrasound can reduce inflammation and aid in the healing of certain tissue injuries. The frequency of the waves used depends on the location of the injury. Higher frequencies penetrate smaller distances, while lower frequencies (more commonly used) penetrate further.

12 Suggest what sort of injury could be treated with ultrasound healing.

Ultrasound can activate white blood cells (leucocytes). High frequency sound waves cause granular (grainy) leucocytes to release histamines. Histamines increase the diameter and permeability of blood vessels. The effect of histamine production is to attract *monocytes* (large white blood cells) to the site, where they clean up any debris that would aggravate the damaged tissue. At the same time, ultrasound activates *fibroblasts*, cells associated with connective tissue. Fibroblasts secrete substances that form collagen, the main component of healthy connective tissue. The combined effect of activating these cells is to speed up healing and reduce inflammation by removing any contaminants and assisting in reconnecting damaged tissue fibres.

13 Research – Find out the chemical formula for histamines.

14 Research – Name two common respiratory reactions in which histamines are produced in the blood.

15 Listed below are three areas of specialisation in ultrasound treatment. Find out what each one is used for.

a vascular

b obstetric

c ophthalmologic

Echocardiography – Cardiac ultrasound

Echocardiography uses ultrasound on the heart muscle to produce images with greater detail than X-rays, but without exposing the patient to radiation. Transducers are placed at the breastbone (sternum) and ribs, using a water-based conducting gel. The transducer is directed towards the heart where it detects the echoes it receives, then relays the echoes to a console. The console uses the information to construct an image. Other names for this technique include Transthoracic echocardiogram and Doppler Ultrasound of the Heart.

16 Suggest why someone might need echocardiography.

Electrocardiography measures the heartbeat by placing electrodes on the patient's skin to detect the electrical activity of the heart. The resulting printout, or *electrocardiogram* (also called EKG or ECG) illustrates the heartbeat ('lubb-dub') activity. The pattern produced is one of wavy lines followed by jumps and slight dips. The peaks and dips represent the *lubb* sound of the heartbeat and the straight lines the *dub*. By comparing the line patterns produced by a healthy heart, physicians can recognise signs of heart problems.

17 What is an electrode?



- 18 How do you think the development of echocardiography and electrocardiography have affected the incidence of heart problems?
-

X-rays

X-rays are used to produce images of the inside of the body. X-rays are high energy electromagnetic radiation, which penetrates the soft tissue of an organism until it hits, and is absorbed by, denser material (such as bone). The image produced resembles a photographic negative. Differences in bone density are indicated by varying shades of black and white. Denser material appears white, with other structures ranging from white to shades of grey. X-rays are not as effective at producing images of soft tissue. The denser the material, the greater the absorption and the lighter the image.

- 19 Why is soft tissue transparent to X-rays? _____
- 20 The gall bladder is a soft organ, but it is often X-rayed. Why? _____

Thermography

Humans emit body heat – this idea is employed in thermography. The word *therm* refers to heat, telling us that thermography is a technology that uses heat to produce an image. Thermography detects the slight differences in temperature found over an area of tissue. These differences appear as distinct shades on the image, producing a visual representation of the tissues being investigated.

- 21 List one advantage of thermography over X-rays.

- 22 List one disadvantage of thermography over X-rays.

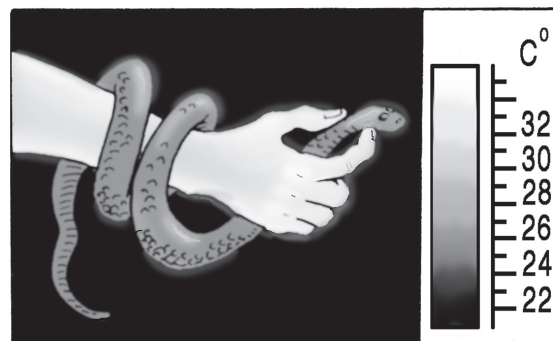


Figure 3

Tomography

Using ultrasound or x-ray technology, photographs of a cross-section of an organ can be produced. *Computer Axial Tomography (CAT scanning)* uses computerised tomography to glide across tissues, detecting slight differences in the shapes and angles of the tissues (*axial* meaning 'around', *tomo* meaning 'section, slice'). Scanning around organs from many angles produces CAT scans. By approaching the organ from different angles, the information is pieced together to produce a three-dimensional image of the scanned tissues.

- 23 What is the difference between two-dimensional images and three-dimensional images?
-

MRI

Our world is made up of atoms and our bodies are no exception. Every type of atom has unique features that allow us to identify it. For example, hydrogen atoms have one proton in their nucleus and are a specific shape; whereas helium atoms contain two protons and are a different size. In the presence of a strong magnetic field, *Magnetic Resonance Image Scanning (MRI)* produces images by measuring the properties of the atoms present in the organs under investigation. Using the unique properties of atoms, and their positions relative to each other in the body, the computer assembles the information into an image of the scanned organ.

- 24 In your workbook, draw up and complete a table to summarise the procedures covered so far in this worksheet. Use these headings: *Name of procedure, What it is used to diagnose, Unique feature (e.g. uses heat)*.

Keyhole surgery: Laparoscopy

This procedure involves making the smallest possible incisions in the patient. Laparoscopic surgery is commonly called keyhole surgery since the scar (if any) resembles a keyhole. Sometimes operations are performed using natural body openings.

- 25 What is an *incision*? _____
-



Using long-handled instruments and tiny cameras (which employ fibre optics), doctors do not look down into an open incision as they do during major invasive techniques. Instead, they use a small screen positioned nearby, to which the camera images are transmitted so the doctor can see what is happening. Patients recover more quickly from keyhole surgery than from standard procedures. Side effects and scarring are also greatly reduced or absent with this technique. Keyhole surgery has been called 'band-aid' surgery, because that is usually all that covers the surgical site after the operation.

- 26 Research** – Endometriosis is suffered by millions of women every year. It is commonly treated using keyhole surgery. Find out what endometriosis is and further details about its treatment.

- 27 Research** – Describe the difference between a 'local anaesthetic' and 'general anaesthetic'.

In the past, the 'tying' of the fallopian tubes involved making an incision of 10–14 cm across the 'bikini line' of a woman's abdomen. The area was opened, muscles and other tissue were cut through, and the fallopian tubes were then clipped (using synthetic clips), stitched or crushed, to prevent the egg from making its way into the uterus for fertilisation every month. Recovery times varied from several days to two weeks before the woman could return home, after which only light activity could be performed for a period of several more weeks. This procedure involved high hospital costs, long periods in hospital and considerable pain to the woman.

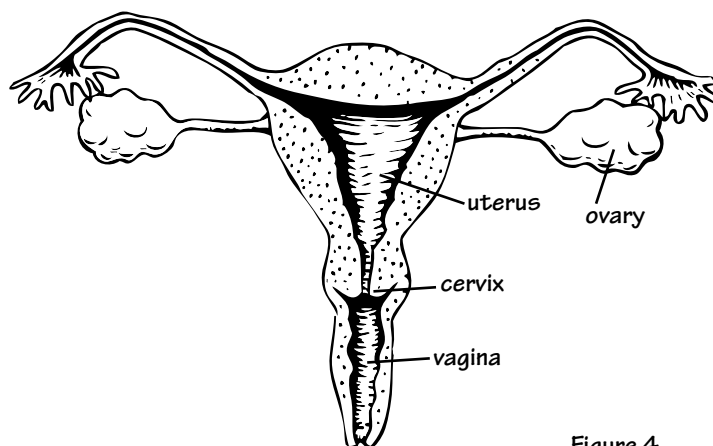


Figure 4

- 28** Label the fallopian tubes on figure 4.

This operation is now commonly performed by keyhole surgery, through two small incisions in the woman's abdomen. The doctor performs the procedure by manipulating long-handled instruments while referring to a camera that shows what is happening inside. The operation takes less time and recovery is much quicker, with the patient usually returning home the same day.

- 29** List the advantages of clamping or 'tying' the fallopian tubes via laparoscopy to the patient, the patient's family, the hospital and the society.

- 30** Tick the following if they are *advantages* of keyhole surgery, and cross them if they are not.

- | | | |
|-------------------------------|---------------------------------|--------------------------------|
| a Quicker to carry out | c Less risk of infection | e High technology |
| b More skill needed | d More rapid healing | f Less obvious scarring |

Life Support Systems use artificial means to keep people alive during operations or while the body repairs itself. Without such systems, many people would die either during operations or later, because the body's resources are stretched too far trying to regain health or repair damage. There is no set life support system, as there is no single condition that requires it. If someone has a severe asthma attack, the system required will be one of maintaining lung function. Another person, whose lungs are functioning properly, may require help keeping his or her heart beating. The equipment involved also helps hospital staff to monitor a patient's condition, particularly when the patients are unable to speak.

- 31** Find out what criteria are used to decide whether or not to keep a person on total life support if they are unconscious, unable to breathe by themselves and are showing no signs of recovery.

- 32 Research** – Find out how the laparoscopy technology was developed.



- 1** Mark the following statements as true or false. Then correct the false statements so that they are true.
- | | |
|---|--|
| a Monocytes are large white blood cells._____ | e Thermography uses heat to produce images. _____ |
| b Another word for white blood cell is leucocyte._____ | f EKGs and ECGs are used to detect brain activity._____ |
| c X-rays produce images of soft tissue._____ | g CAT scans scan organs from many angles. _____ |
| d Cardiac ultrasound uses echoes to produce images._____ | |
- 2** Write brief answers to the following.
- a** What is a transducer and what is it used for? _____

- b** What does MRI stand for? How does MRI produce an image? _____

- c** Discuss the difference between invasive procedures and minimally invasive procedures. _____

- d** What is echocardiography and how does it work? _____

- e** How can ultrasound be used for healing tissue? _____

- 3** Write each area of ultrasound specialisation beside the correct description: *vascular, obstetric, ophthalmic, sonography*.
- a** To do with pregnancy and birth. _____
- b** To do with diagnosis using ultrasound. _____
- c** To do with blood vessels or other fluid transporting vessels. _____
- d** To do with the eye. _____
- 4** Describe one of the non-invasive diagnostic methods you have studied in this unit.

- 5** What is an electrode? Why are they used? Would they be considered invasive?

- 6** What is considered more dangerous in the long term, ultrasound or X-ray? Why?

- 7** Would x-rays ever be useful in diagnosing soft organ problems? Explain.

- 8** Cross out the incorrect words in *italics*.
- a** CAT scan is an acronym for computer axial *topography* / *thermography* / *tomography*.
- b** MRI uses recognisable features of *atoms* / *tissues* / *electrical impulses*.
- c** *Laparoscopy* / *bandaid* / *bioscopy* is not another name for keyhole surgery.
- d** Thermography is used to detect differences in *size* / *heat* / *chemical composition*.
- e** CAT scans do not look for differences in *angle* / *shape* / *density*.
- f** A hysterectomy is considered *non-invasive* / *minimally invasive* / *invasive*.

How far is too far?

In biomedical research, ethics (moral values) must be considered. Ethics differ between individuals and groups of individuals. What one considers acceptable, another does not. Many cultures and religions do not approve of ‘tampering with nature’, often arguing that humans were not designed to extend life unnaturally. Others argue that great minds are meant to be used – hence, if we can find a way to do it, then we should do it! No matter how far we progress with scientific research, ethical issues will always be nearby to remind us of other considerations involved with developing technology.

Animal Ethics committees are important groups that represent the welfare of animals. In the past, the welfare of animals in scientific research was not considered important, and many animals suffered horribly. Today, people generally respect the lives (and feelings) of animal subjects used for experimental purposes. Ethics committees are often made up of a variety of people, for example, a teacher, a clergyman, several scientists and a police officer. This is to ensure a ‘public’ opinion is reached.

- 1 Why isn’t the committee made up entirely of scientists? _____

- 2 What sort of questions might be asked of scientists looking for approval to work with animal subjects? Why? _____

- 3 ‘Right to life’ campaigners believe that people do not have the right to use medical technology to take their own lives (humanely) when they are terminally ill. Discuss the ethics of this ‘right’, including reasons for and against.

‘Stolen body parts’

In mid-2006, newspapers reported that American firms had unknowingly been using products made from stolen body parts. Bones, ligaments, skin and other tissues were removed from corpses in funeral homes, without the knowledge of the relatives of the deceased. The body parts were sold onto other firms who used them as raw materials in making other products that were used to treat many complaints. The US Food and Drug Administration ordered a recall of the products.

- 4 Comment on the ethical issues involved in the paragraph above. Discuss the roles of the initial provider and the user of the final product.

- 5 **Research** – Australia has a body to oversee health matters such as those described above. What is its name?

Autopsies for research

The *Spanish influenza* epidemic lasted from 1918 to 1919 and killed over 12 million people. Recently scientists have been exhuming (digging up the remains of) people who died of this disease to compare with current strains of ‘flu, especially the H5N1 variety.

- 6 Is it right to exhume dead bodies simply for medical interest? List arguments for and against.



In 1998, the world’s first successful hand transplant was performed (using the hand of a man who died in a motorcycle accident). An Australian microsurgeon was one of the team of specialists involved in this ground-breaking operation.

The transplant required the usual treatments of immuno-suppressing drugs and other therapies. Immuno-suppressing drugs restrain (suppress) the immune system so that it doesn’t recognise the ‘non-self’ tissue that is transplanted and so doesn’t attack it. This way, the body (often) accepts the ‘foreign’ tissue. However, despite the procedure being a technical success, no one foresaw the recipient’s emotional reaction to the surgery. This led to the receiver failing to take his medicine and his body completely rejected the transplanted hand.

1 The immediate reaction of many people upon hearing this story is surprise that someone who had lost a hand wouldn’t be thrilled to have a functional ‘living’ replacement. What sort of emotional problems do you think the recipient experienced?

2 For many years, science fiction stories have discussed the use of cryogenics to ‘deep freeze’ bodies of people who are suffering from incurable illness, planning to revive them once a cure is found. List arguments for and against the use of this technology.

The artificial heart – A brief history

As early as the 1950s, researchers have worked on developing prototypes to replace a diseased heart. This device would have to be comfortable, not too large and constructed of materials unlikely to cause blood clotting. Since it is preferable that a patient does not undergo multiple operations, the device should also require little maintenance. The heart must also be able to respond to the requirements of the recipient, speeding up and slowing down as needed and be fully contained (sealed inside) to reduce the risk of infection.

Such devices now exist, but their evolution, which continues, has been long and difficult for everyone involved. So far, mechanical hearts have only been temporary solutions, but there is hope that they will eventually be a solution to donor shortages.

3 Draw and label a simple diagram of a mechanical heart in the box at the right. Consider how it would connect to the body’s natural blood circulation system, and indicate the direction of blood flow.

The first major breakthrough came from the work of Dr Robert Jarvik. The *Jarvik-7* artificial heart, made of plastic, aluminium and Dacron polyester, used two pumps to act as ventricles. Despite a long history of application and development problems, followed by many disappointments, *Jarvik-7* was a milestone in directing the evolution of artificial heart design.

4 Suggest potential problems arising from a patient being connected to a machine that pumps their blood.

5 List arguments for and against organ donation. You could conduct a class debate on this issue.



One problem with *Jarvik-7* was the necessity of wiring its cumbersome machinery. This forced recipient to remain near the console, with the added inconvenience of having assorted wires and rubber tubing sticking out of the chest cavity, increasing the risk of internal infection. In December 1982, an American dentist named Barney Clark received a *Jarvik-7* artificial heart and survived for 112 more days. A second recipient, Thomas Gaidosh of Pennsylvania, had the same model implanted for several days until a donor heart was found and transplanted. William Schroeder, who died in August 1986, holds the *Jarvik-7* survival record (620 days). Mr Schroeder grew to loathe the huge console, to which he was forever attached and considered himself to be the 'living dead'. This general opinion earned the *Jarvik* devices the name '*Dracula machines*'.

- 6 Why do you think William Schroeder considered himself to be 'the living dead'? _____
- 7 List the emotional factors for recipients that people who develop devices like the artificial heart should consider. _____

AbioCor

In August 2001, the world watched with fascination a television interview with 59-year-old Robert Tools, the only living person without a heartbeat! A month earlier, Mr Tools received the world's first self-contained artificial heart, the *AbioCor* model. Unlike the *Jarvik-7*, *AbioCor* functions independently of cumbersome machinery. Mr Tools good-naturedly told the interviewer that he was still getting used to having 'a whirring sound' instead of a heartbeat. Two weeks after receiving the implant, Mr Tools developed an infection and fever, though the *AbioCor* continued to function properly. Mr Tools lived 151 days with his implanted heart, dying from multiple organ failure (commonly caused by an over-reacting immune system) and internal bleeding.

- 8 Do you think it was worthwhile for Mr Tools and Barney Clark earlier to have undergone their heart operations, only to have lived 151 and 112 days respectively? Explain your answer. _____
- 9 Would you choose to have an artificial heart if you were in this situation? Explain. _____
- 10 Study the diagram of the *AbioCor* artificial heart (figure 1). Describe how it works and outline how it differs from the *Jarvik-7*. You may have to do some research. _____

At present, artificial heart operations cost about half that of a heart transplant. When an effective, long-term device is developed, it is expected that increased production will further reduce the cost.

Using the principles of tiny turbine-containing devices, other artificial heart models are currently being developed, including the *BeneCor* and the battery-sized *Jarvik 2000*. Doctors hope that in the future, patients with artificial hearts will live nearly normal lives. At present, the device is still considered a temporary solution, used to gain time while other solutions are investigated. Thanks to the courage and determination of researchers and the brave pioneers who tested them, we come ever closer to producing long-term, permanent devices.

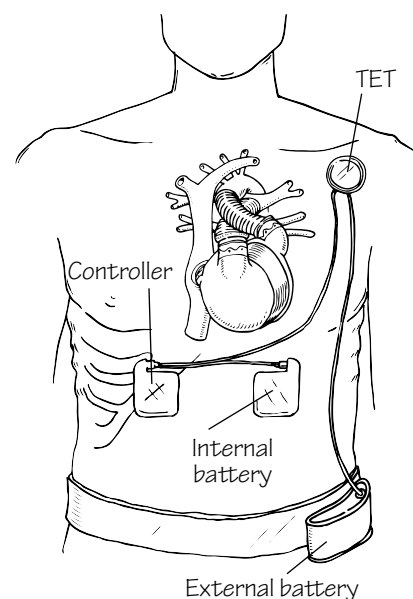


Figure 1



Many Australians love ‘footy’, be it AFL, soccer, rugby league or rugby union. People who don’t love it or play it still probably know someone who does. The toughness, stamina and downright ‘guts’ of the players can be admirable, and scouts are always on the lookout for new talent. These days, a new form of scouting has been proposed – and it is having a great impact.



- 1 Choose one code of football (or another sport) and outline the characteristics that would make a person successful at this sport.

The practice of using genetic tests to determine the likelihood of an individual being a good player has been proposed for Australian football leagues. This is highly controversial, and raises many ethical questions. It renews the debate of the effects of nature versus nurture in determining what a person will become. Due to heredity (genetic make-up), we are born with certain characteristics – such as how tall we might become or what colour our eyes will be. People in favour of DNA testing believe that the ability to identify certain characteristics will help to produce superior athletes. However, nurture (how you are raised, your life experiences) also plays a very important role in what people become.

- 2 List two examples of types of *nurture* of a child.

- 3 Consider a hypothetical set of identical twins, separated at birth. Their DNA will be exactly the same, so only their life experiences will distinguish the attributes they possess. If one twin grew up in poverty and had abusive parents, and the other twin grew up wanting for nothing in terms of love, food and opportunities, how might their sporting skills be affected? Would their skills be very similar? Very different? Explain.

- 4 Which do you believe is more important – nature or nurture? Or do both have an equal effect? Give reasons for your answer.

- 5 If you read the sports pages of a newspaper, you will probably see the terms *attitude*, *motivation* and *character*. Briefly define each of these terms.

- 6 Do you think DNA testing could show the likelihood that players will demonstrate any or all of the qualities mentioned in the previous question? Explain.

- 7 A prominent football coach was concerned that DNA testing would lead to searching for athletes, rather than football players. Football players *are* athletes, so what did she mean?

ACTN 3 Gene

Every school seems to have natural athletes who finish first in races or outplay the rest of the team in sports. In 2003–04, a correlation was made between the presence of a certain (R–) form of the ACTN 3 gene and athletic performance. This led to people suggesting that DNA testing could be used to find out if potential football players had this gene – positive results should ‘ensure’ a good player. However, a good athlete is made up of more than a single gene. Most of the genes in our DNA work with others in a linkage relationship. Therefore, although they haven’t yet been identified, other genes are probably also involved in making a good athlete. But there’s more to an athlete than his genes. Good athletes are motivated to succeed. They are willing to ‘do the hard yards’ at training, follow a certain diet and push themselves as far as they can go.



Extracting DNA for testing is quite simple. A cotton swab is rubbed inside the cheek and the genetic material is extracted from the (invisible) cheek cells that are wiped onto the cotton. Even if DNA testing could reveal the *potential* for certain characteristics, it cannot guarantee that a certain football player will be a good investment for a team. DNA isn't the only influence on what happens on or off the field.

- 8** Just because an individual (from a DNA perspective) looks promising, they might not make the grade. Outline possible reasons for this.

There are other factors that must be considered in regard to DNA testing. If people are identified as having the potential to be good football players, this might discourage them from considering other careers for which they are also well suited. Discrimination is another consideration. DNA testing could be used to suggest strengths but it can also be used to indicate weaknesses and predispositions to certain medical conditions. These conditions may never materialise, but being told about the increased risk can affect how management choose to place (or not place) individuals in sporting positions.

- 9** Consider a 16-year-old girl, who is told that she carries a gene that makes it more likely she will develop a life-threatening disease in the future. Outline how the girl might be affected by this knowledge.

Those in favour of DNA testing argue that by knowing a person's genetic make-up, training programs can be tailored to suit their strengths and weaknesses, and also enable the management to place the individual in the position that genetics suggests is best for them. However, players may not wish to play the sports their genes suggest they should.

- 10** Explain the difference between *mental strength* and *physical strength*. Could one affect the other? If so, how?

- 11** What is some of the *scientific* information DNA testing could reveal? _____

- 12** Ethics involves moral issues. Discuss the ethical issues involved in genetically testing football players.

- 13** Do you agree or disagree with DNA testing football players? Why? _____

- 14** Do you think DNA testing would be a more effective way of scouting for new talent? Explain.

Prenatal recruitment: An example of an ethical issue

Imagine that on April Fool's Day, a newspaper columnist 'revealed' the following (spoof) story:

'Pre-natal selection. DNA scouts acting with management and club doctors have offered current and former footy greats contracts to have children with selected females to produce male offspring with the ideal genes to be great footy players. Sums in excess of \$2 million have been mentioned, plus medical benefits. Some rival teams are rumoured to be planning legal challenges against the move and others have similar programs in the planning stage.'

- 15** Comment on the moral and ethical issues that would be involved if this story were true.

- 16** What practical problems might this arrangement involve, if it were true? _____

- 17** Compare the approach described in the newspaper article with traditional scouting. _____



The term *cyborg* or *cybernetic organism* conjures up images of humanoid robots with super-human abilities, commonly found in science fiction. Cybernetics is the science of control and communications in animals and machines. A cyborg is a 'blend' of human and machine. The idea of attaching artificial parts to humans modernised the Frankenstein story.

- 1 What is meant by *humanoid*? _____
- 2 Suggest why science-fiction writers have often made their characters cybernetic organisms rather than 'ordinary' humans.

- 3 List two other fictional cyborgs, and describe how close current technology is to making them possible.

- 4 Draw and describe a cyborg in your workbook.

The field of *Bionics* is concerned with developing engineering solutions for impaired (damaged or destroyed) body parts. It specialises in creating electrically operated devices to replace or enhance the functions of body parts.

- 5 List six bionic devices and the body part(s) each one repairs or assists.

- 6 Name a high-profile person who uses (or used) a bionic device. _____

Biomaterials and bionic devices

Advances in science and technology have increased our ability to improve, enhance and prolong life using bionic devices. By adopting a multidisciplinary approach, *biomaterials* (materials developed to replace living tissue) have been formed to create *biomedical* devices. A biomedical device is a piece of equipment used for a medical purpose, such as helping a person to walk or ensuring a heartbeat remains steady (pacemaker). When biomedical devices are implanted within the body, as a pacemaker is, they are called *implants*. These *prosthetic* devices have revolutionised our approach to illness, injury and disability by taking over, at least in part, the work performed by a damaged or destroyed body part.

- 7 What do the following define?
 - a A device designed to correct an anatomical problem or to create a more desirable appearance that is placed in the human body by surgery. _____
 - b A synthetic material that is used to replace part of an organism or to function with living tissue.

 - c An artificial device used to replace a body part. _____
 - d A living being who is a composite (mix) of both living and electrical/mechanical artificial body parts.

Biomaterials are manufactured materials designed to replace living tissue. Since many devices are implanted in the body, they require certain properties if they are to be successful. Biomaterials should be strong and must not initiate immune system reactions. Early biomaterials often led to rupturing red blood cells, a dangerous side-effect that has since been addressed. Corrosion and subsequent poisoning also occurred prior to the development of certain metal alloys (around the 1930s).

- 8 What is the 'normal' role of the immune system? _____
- 9 Why is it important that biomaterials be unreactive? _____
- 10 What is involved in corrosion? _____



11 Why must the immune system be considered when implanting biomedical devices?

12 Describe the difference between *biomedical devices* and *biomaterials*.

13 Research – What is an alloy? Find out the names of three alloys. For each one, state what metals are present, list its properties and describe one of its uses.

Blood clots are solid clumps of blood. They can cause heart attacks or strokes. There is always a risk that blood clots will develop as a result of implanting any device, and many improvements have been made to reduce this risk.

14 It is important that the properties of biomaterials feature these characteristics: *biocompatibility, durable, low friction, unreactive, lightweight, strong*. Write each characteristic beside its meaning or purpose.

a The ability to be compatible with and not rejected by human tissue.

b Does not 'rub', for ease of movement and comfort.

c Able to withstand pressures placed on it by the body.

d Does not wear down or wear out easily.

e Won't react with, or be affected by, a body's biochemical reactions.

f Not heavy, avoids making devices cumbersome to wear or use.

Biomedical science involves the interaction between science and technology. Consider the pacemaker, a device used to stimulate and normalise the muscles of an otherwise defective heart.

15 Outline how different branches of science and technology combined to produce the pacemaker.

Aerospace engineering was also involved in the development of the artificial heart.

16 What is the aerospace industry?

17 Name two types of aerospace materials and describe how they are used.

Biocompatibility and immune system response

One of the greatest difficulties faced when transplanting tissue from one individual to another, is the reaction of the recipient's immune system. All living cells contain surface proteins that are unique to the individual (identical twins have identical proteins). The body's immune system detects the arrival of foreign proteins by reading the 'bumps' on their outer coats. It uses this ability to recognise invading bacteria and viruses. When a 'non-self' protein is detected, the immune response is to destroy the invader. It has no way of knowing that the non-self proteins are intended as replacement tissue. Similarly, other materials can initiate attack responses when the immune system recognises their surface structure as non-self. The unreactivity of biomaterials allows them to be largely ignored by our natural immune systems.

18 Rewrite in order of increasing complexity: *organ, cell, organ system, tissue, body*.

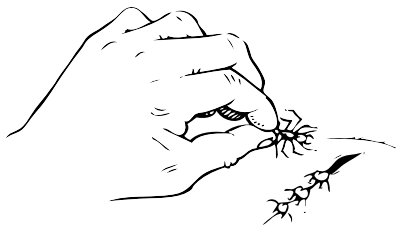
19 Why does the body expend energy to have the immune system 'read' proteins it encounters as they circulate?



Examples of people’s attempts to produce replacement materials for impaired or lost body parts are scattered throughout history.

1 What is meant by *impaired*? _____

Early Egyptians, Greeks and Romans all left evidence (artefacts) that illustrates their use of artificial limbs, including wooden noses! Silver and gold wires, bones, clay, skins and wood have all been used to repair injuries. Mummified remains dating back to 1100 BC (Egypt) have been identified as having stitches (sutures) in the body tissues, while plaster casts date back to at least 2000 BC. Many South American tribes took advantage of the large, sharp pincers found on soldier ants to ‘stitch’ surface wounds. The ants were placed on the open skin and became defensive, pinching the lips of the wound closed in an attempt to attack. Once the pincers were securely in place, the ant was pulled away and its pincers remained in the skin.



Society’s reawakened interest in ‘natural remedies’ has recognised what American Indians, Australian Aborigines and ancient Egyptians have long known about the value of using honey to disinfect wounds. Evidence suggests that these people also used animal skins and/or clay mixtures to form casts. All of these things are early examples of biomaterials.

2 Why might a cast be used? What is its purpose? _____

3 Find out three therapeutic devices or treatments used today that were invented more than 1000 years ago.

How biomedical devices help to improve quality of life

It is always better to possess a healthy organ or limb, than a prosthetic device, because the device is never as effective as the real thing. However, when we lose a functioning body part, prosthetic devices allow people to recover at least some of the ability that was lost. Artificial body parts are called *prostheses* (singular *prosthesis*). Examples include artificial legs, arms, eyes (including lenses), hearing aids and teeth.

Another important quality of these devices is that they can restore human dignity, and some or all of a person’s independence. Most people know at least one person who wears a *crown* over a dead or broken tooth or a whole set (or ‘partial plate’) of *dentures*, both of which are examples of biomedical devices.

4 Inserting an artificial breast after a woman has had a mastectomy due to breast cancer is one example of using cosmetic surgery to improve a person’s quality of life. List two more examples.

5 What is another common term for *dentures*? _____

6 Apart from dentures, name three examples of prosthetic devices.

Using technology derived from digital cameras, the creation of bionic eyes may make the non-functional glass eye a rarity in future years. Less obvious are artificial heart valves and shunts, pacemakers, plates, pins, screws, synthetic arteries, contact lenses and cochlear implants (*cochlea*, meaning ‘spiral hollow’, referring to the inner ear). While many devices, such as spectacles, still resemble early versions, artificial body parts have come a long way, to the point of requiring new terms to describe them.

7 Suggest why a person would have a non-functional glass eye implanted.

8 The cochlear implant has helped many hearing impaired people. The development of a similar device to aid or replace damaged eyes is still many years away. Suggest possible reasons for this.

- 9 Identify the terms described below. Choose from: *artefact*, *bionic*, *cosmetic*.
- a Done for the sake of personal appearance. _____
 - b Electronic and/or biomechanical devices used in conjunction with the human body. _____
 - c An object made by humans remaining from a particular historical period. _____



Microchip is a very familiar word used in bionics. Originally designed for computers, these tiny devices are now also used in home appliances, video cameras and televisions, and many other items. Even the family pet can have a microchip implanted (at the back of the neck) to provide ownership details in case they are lost. Microchips are also used in the operation of biomedical devices.

- 1 List advantages and disadvantages of using a microchip implant in a pet rather than a standard tag on a collar.

- 2 List two other examples of how microchips are used.

Another familiar word used in bionics is *laser*. We've all heard of laser printers, laser copiers, laser surgery and, for the science fiction fans, laser beams! The word laser is an acronym (a word formed by the first letter of a number of words) for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation. Lasers can direct extremely fine beams of electromagnetic radiation of a specific frequency, with great precision and intensity. Laser technology is commonly used in minimally invasive surgery. A laser beam can concentrate a lot of energy onto a very small area.

- 3 Laser beams are also used in CD and DVD players, and in very cheap devices such as laser pointers. Explain why a person using a laser pointer to draw attention to items displayed would need to be careful.

- 4 Lasers emit a beam of *monochromatic light*. Circle the correct definition of monochromatic light:

- a Light that only goes on once.
- b Light that damages chromosomes.
- c Light that is mainly of one colour.
- d Light that can only be seen in darkness.

An *actuator* is probably the least familiar of the bionic components, yet the most easily understood. To *actuate* means to make something move a certain way. Actuators, to put it simply, make things work. Actuators put devices into action using motors, switches or whatever else is required.

- 5 Could the starter on a car be considered an actuator? Explain.

- 6 List another example of an actuator.

Biomedical materials – more detail

Polymers – Synthetic polymers are an important component of many biomedical devices. When large numbers of identical molecules (sub-units or 'monomers') bond together to form a very large molecule the new, larger, molecule is called a *polymer*. Polymers, more commonly known as plastics, are all around us. The average home (and wardrobe) contains many items made from polymers. The reaction used to form polymers is called *polymerisation*.

Figure 1 shows a simplified diagram of a section of a polymer chain, including individual monomers and the chemical bonds that hold the monomers together.

- 7 Referring to figure 1:

- a Identify part A.
- b Identify part B.
- c Identify part C.

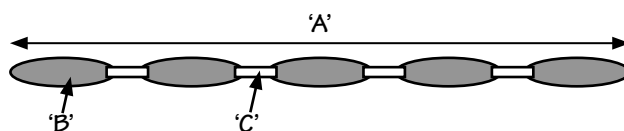


Figure 1

Polymers can be designed to be unreactive and have low friction properties. This makes them versatile substances with which to design prosthetic devices. Many polymers are waterproof and possess elastic properties which allow for increased flexibility.

- 8 What is meant by *elastic*?

Their lack of reactivity makes them suitable to implant into human bodies, which are a hive of biochemical reactions, and would quickly reject them otherwise.



Teflon (PTFE or polytetrafluoroethylene) is an easily shaped, slippery, plastic material, best known for its use in coating ‘non-stick’ (low-friction) pots and pans. PTFE is a versatile material used to make artificial heart valves (among other things).

9 Outline some properties not already mentioned about PTFE, that it would be likely to possess, in order to be suitable as a heart valve replacement.

10 Figure 2 shows a sectioned view of the heart.

- a Indicate the location of heart valves on this diagram.
- b Show the direction in which blood flows through these valves.

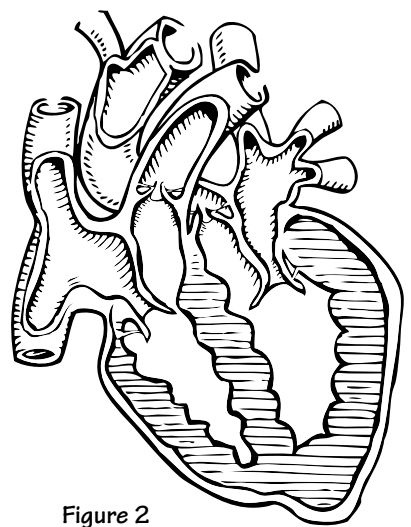


Figure 2

Polyethylene is a thermoplastic polymer often used to cover the articulating (jointed) ends of artificial joints. It is also useful for creating artificial blood vessels and body parts which are normally made of cartilage (e.g. nose), although ultra high molecular weight polyethylene (UHMWPE) is most commonly used as a cartilage replacement.

11 What is the role of cartilage in an adult human skeleton?

12 List five examples of polymer-based objects in your home.

Although people only developed polymers in the 20th century, nature has been creating polymers for millions of years. Commonly used natural polymers include wool and silk.

13 List two uses each for wool and silk.

14 Name the organism that makes wool.

15 Name the organism that makes silk.

Pyrolytic carbon (PyC) is a durable polymer that resembles graphite. One of its main applications is to coat other materials that are used to make biomedical devices. In addition to durability, pyrolytic carbon is *thromboresistant* – this means it will not cause the blood to clot when implanted in the body (*thrombus*, meaning ‘clot’). Many artificial heart valves are coated with pyrolytic carbon, making them safe to implant. Pyrolytic carbon is formed by heating rubber in the absence of oxygen, to break it down (*pyrolysis*, Greek *pyros* meaning ‘fire’ and *lysis* meaning ‘to break apart’).

Ultra high molecular weight polyethylene (UHMWPE) is a durable, biologically compatible, prosthetic alternative to cartilage that is used for surrounding heavily used ball and socket joints. This polymer can withstand friction at the site. It can also tolerate high temperatures and does not change shape over time.

16 Identify the terms described below. Choose from: *compatible*, *clot*, *durable*, *polymer*, *pyrolysis*, *versatile*.

- a Lasts a long time _____
- b Has many uses _____
- c Substance made of many repeating parts _____
- d Derived from the Greek term, thrombus _____
- e Works well with other systems and materials _____
- f Chemical change caused by (extreme) heat _____

Silicones – These are synthetic polymers based on the element silicon. If you wear glasses, you may have noticed the soft nose pads – these are usually made of silicone. Silicones are useful in creating prosthetics because they are non-toxic and come in many different forms. A gelatinous form can be implanted as artificial breast tissue after a mastectomy (removal of a cancerous breast) or for cosmetic reasons. The more solid forms are used as outer coatings for bionic devices, or as insulating materials in other devices. Silicone is used to replace or reshape pieces of ears, chins and other body parts. It is also used to coat and insulate electrodes such as those used in pacemakers.



17 What is meant by *cosmetic*? _____

An important difference! It is important to note the difference between *silicon* and *silicone*. Silicon is a naturally occurring semi-metal (metalloid) element. Its semi-conducting properties make it useful in the manufacture of microchips. Silicones are durable synthetic polymers often used to make polishes, lubricants, electrical insulators and now, biomaterials.

Figure 3 shows a simplified part of the structure of a type of silicone. R_1 and R_2 represent carbon-based groups.

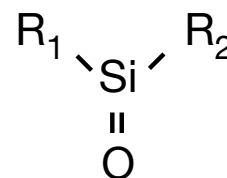


Figure 3

18 Which element is represented by Si? _____

19 Which element is represented by O? _____

Other materials used to make biodevices contain *carbon fibres* and *superalloys*. As the name indicates, carbon fibres are carbon-based fibres. The advantage of carbon fibre is its great strength and relatively light weight (low density).

20 Figure 4 shows a simplified version of the structure of carbon fibre.

Which element is represented by the letter 'C'? _____

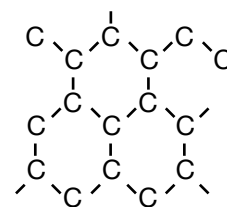


Figure 4

Alloys are metals made by combining at least two elements at least one of which is metallic. One or both elements may be inferior on their own, but are combined to produce a new metal with a desired quality or property.

21 What is meant by *inferior*? _____

Common properties of alloys include increased strength, texture, and ability to withstand erosion. (Stainless steel is a good example.) A *superalloy* is a superior alloy, excellent for use in the creation of artificial joints, pins, plates and other prosthetic devices. In addition to strength, superalloys are unreactive. Titanium, a metal often used in aircraft and rocket engines, is a corrosion-resistant metal (resembling steel/aluminium) used to make pins, screws and other fixation devices.

22 Outline potential problems associated with using metals to connect bones or make artificial joints.

Prosthetic cements – (In this context, *cement* refers to a glue, not the building material.) While cement is commonly used in dentistry, there is a preference for avoiding the use of cement on internal implants where possible. However, with many devices it is better to cement them to the patient's bone if they cannot be securely fastened using pins and screws alone. Devices can later be uncemented if necessary. Much research is currently being undertaken to develop new bone cements. A great difficulty with current cements is that they do not always remain attached to the bone. Bone resorption, when the bone absorbs some of the cement, often occurs. This leads to gaps that loosen the bond and deform, or reshape, the bone. Shrinkage and break down are also a problem. For a cement to be useful, it must be strong, stick (adhere) well, and not shrink.

23 Suggest problems that might arise from bone resorption. _____

Adherence involves a successful *interface* at two sites. An interface is where two faces, or boundaries, meet. These sites are:

- the *device–cement* interface (the boundary where the cement sticks to the device)
- the *cement–bone* interface (the boundary where the cement sticks to the bone).

If interface fails at either site, the device will loosen and require re-cementing and/or repair. Bone remodelling can also occur, changing the bone's original shape and creating further difficulties. This sort of bone erosion is called *osteolysis* (*osteo* refers to bone, *lysis* meaning 'to break apart').

24 Suggest why the use of cement is avoided where possible with implants, but not in dentistry.

The most commonly used medical and dental cement is known as PMMA (polymethylmethacrylate). Its trademark name is Perspex. It is a clear polymer and its major use is as a replacement for glass. Perspex is strong, easily cut into shapes, painted and cemented.

PMMA can also be used as a cement for attaching bone tissue. Not all implants require cementing. Those that do not are designed to have natural bone tissue grow around them, so that with pins and screws, they fit more securely.



25 List properties of Perspex that are likely to make it suitable as a cement for bones.

26 List three uses of Perspex, other than as a cement. For each use, identify the properties of Perspex that make it suitable.

Pins, screws and plates are designed to connect parts of the skeleton. Screws are most frequently used, followed by pins and then plates. These devices are usually made from stainless steel or other superior metal alloys, but some are made of bioglass-metal fibres and other similar materials. Screws are used to attach plates, and pins attach bone in places that are hard to reach, where plates cannot fit.

27 Suggest a part of the skeleton that would be hard to reach, requiring a pin rather than a plate.

28 **Research** – Find out the names of three other alloys and identify the metals present in each one.

The majority of these fixation devices are used on long bone fractures, but not all. Plates are flat structures that are commonly used to cover badly fractured bones. Screws are used to attach them firmly into place, and they can require cementing.

Plates can also have objects attached to them, such as false teeth. The denture plate is made of acrylic or metal, and fits along the roof of the mouth. It has false teeth attached around its rim. Plates, usually titanium (or titanium alloys), have also been implanted in the skull to protect the brain where skull damage or deformation has occurred.

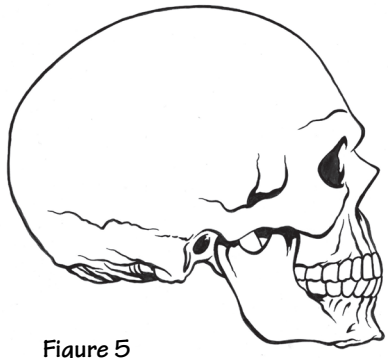


Figure 5

29 Suggest why plates would be used where skull damage has occurred.

Until unreactive metal alloys were developed, pins, screws and plates could not be used inside the human body. Most metals react with body fluids, often poisoning the patient. The new metal alloys are also much stronger and better at withstanding the stresses placed on them by body weight. Earlier metals lacked this strength, often becoming brittle after repeated use.

30 List possible effects of a plate or pin becoming brittle.

Crowns and dentures – Crowns are often called ‘caps’ because they are worn like a cap over a broken or dead tooth. Crowns look like real teeth, but are not a true false tooth. They are often used in the case of a root canal operation, where decay has destroyed most of the living part of the tooth so a dentist must fill it with cement. The loss of blood supply to the tooth eventually causes it to die, become brittle and ultimately shatter. The crown is cemented over the affected tooth, encasing it to prevent further damage, including that caused by other teeth shifting as a result of the gap produced by a lost tooth. In the case of a live but broken tooth, a crown masks the damage. Whole sets of artificial teeth are called dentures. False teeth are traditionally made of porcelain to resemble the calcium-based, enamel outer-coating of real teeth. However, many now incorporate metals with the ceramic to reduce the possibility of breakage, or are made of acrylic materials.

31 Describe the difference between dentures and crowns.

32 Some people with perfectly healthy teeth may want crowns. Suggest a reason for this.



Any artificial device that is designed to replace a living body appendage is called a prosthetic limb. The limbs they can replace include arms, legs, feet and hands.

- 1 Define *appendage*. _____

Wooden crutches, ‘peg’ legs and metal ‘hand’ hooks date back to the mid 1700s, and are early examples of prosthetic limbs. Peg legs replaced lost legs, and to stop bleeding, arteries were severed after the leg was removed. In the case of bleeding arteries, the peg was often screwed into the artery, a painful and dangerous practice. Other peg legs were removable and attached by straps. Today, depending on the cause of limb loss, some prosthetic limbs can be connected to undamaged muscles, and the muscle’s natural ‘circuitry’ used to operate the *myoelectric* (*myo*, meaning ‘muscle’) prosthetic limb.

- 2 Suggest why using a peg leg to seal an artery could be dangerous. _____

- 3 Apart from working in a similar manner to the missing limb, list other factors that must be considered in manufacturing an artificial limb such as a leg, hand or arm. Consider cosmetic reasons – why might appearances matter?

- 4 Are crutches considered to be prosthetic devices? Explain your answer.

Shunts and valves – To shunt something means to divert, or redirect it. Shunts are tiny tubes that are used to bypass weak, damaged or blocked areas. In the case of living organisms, these areas are usually veins or arteries. For example, if a blood vessel has been damaged, a shunt can be inserted to allow blood flow to continue and prevent the vessel from rupturing or remaining blocked.

- 5 What might happen if a blood vessel ruptured? _____

Lenses – Eye glasses (spectacles) and contact lenses are devices used to correct vision impairment. Lenses come in a variety of shapes and curvatures, designed to correct the defect in the patient’s eye.

- 6 Describe, with the aid of simple diagrams, the major differences between spectacles and contact lenses. List advantages and disadvantages for each.

Faulty heart valves – A ‘crisp click’ describes the sound made when one or both heart valves bulge during systole (contraction of the heart muscle). The terms ‘honks’ and ‘whoops’ are also used to describe the sounds made by faulty heart valves. Each different sound helps physicians to determine the cause of the problem. Some valves degenerate, while others can prolapse, or sag. Warning signs for faulty valves include chest pains, palpitations, dizziness, migraines and fatigue.

- 7 What does *degenerate* mean? _____

- 8 Explain why chest pains, palpitations, dizziness, migraines and fatigue can indicate faults with heart valves.

Artificial valves – Leaky or damaged valves are often detected (using a stethoscope) by the hissing sounds they make. The hiss is made by the noise of the backward flow of blood (backwash). Consequently, this backflow reduces



the amount of blood that reaches certain organs. The heart may then pump harder, in an effort to compensate for the shortage, and this places the heart under additional strain. The heart itself requires a regular flow of fresh blood, and faulty heart valves can lead to pooling and subsequent clotting of 'stale' blood in the ventricles. The first artificial valves were implanted in the 1960s. They wore out after 2–3 years, and could be heard by the implant recipients. Today's artificial replacement valves fall into two general categories: originating from living tissue or entirely artificial.

Fully artificial heart valves contain no living tissue and fall into two categories: biomaterial and mechanical valves. Biomaterials imitate real tissue. Mechanical valves are often titanium alloys or stainless steel. Many contain biosynthetic coatings to prevent blood clotting. The tiniest edge can activate the body's platelets, which then clump to form internal 'scab' seals. For this reason, people with mechanical valves require anticoagulants (medication used to prevent blood from clotting). Entirely biosynthetic valves do not usually have this effect.

9 What considerations are involved when using living tissue to make artificial valves? _____

10 What are the risks of long-term use of anticoagulants? _____

11 Why is a bleeding artery more dangerous than a bleeding vein? _____

12 **Research** – Draw up a class list of types of biomedical devices. Work in pairs or small groups. Allocate one device to each group. Collect detailed information about the device, including its use, development, the materials from which it is made, and the advantages and disadvantages of using it.

13 Mark the following statements as true or false. Then correct the false statements so that they are true.

a Chest pain, palpitations and dizziness can be a sign of faulty valves. _____

b Out of pins, screws and plate, pins are the most frequently used. _____

c Metal is the main component of fibre optics. _____

d Pyrolytic carbon is formed by heating rubber. _____

e Mechanical valves contain some metal. _____

f Perspex™ is used as a medical and dental cement. _____

g Silicone is a naturally occurring metalloid element. _____

h Evidence suggests that prosthetic devices have been used for centuries. _____

i Once cemented, biomedical devices cannot be un-cemented. _____

j Fibrinogen is a blood protein involved in clotting. _____

k Pins, plates and screws are made of metal alloys. _____

14 Answer the questions below then use the boxed letters to complete the definition. (Some letters are used more than once.)

Bionics: _____

a To combine several disciplines. ____ [] _____

b Biomaterials are designed to replace living what? ____ [] ____

c Another word for rust. ____ [] ____ [] ____

d A blend of metals. [] ____

e Term used for 'beauty' uses of biomedical devices. ____ [] _____

f To be medically inserted in the body. ____ [] ____ []

g Common name for polytetrafluoroethylene (cartilage replacement). ____ [] ____ []

h Wool and silk are examples of naturally occurring what? ____ [] ____

i Another word for 'stitches'. ____ [] ____ [] ____

j Greeks, Romans and early ____ [] ____ used prosthetic devices.

k Latin word, meaning 'to clot'. ____ [] ____



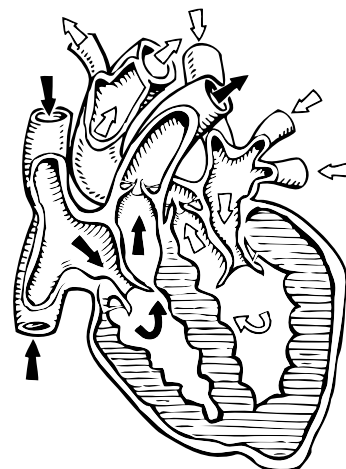
Cardiovascular system: heart and blood vessels, organs involved in the circulation of blood around the body

Heart disease can develop in different ways. Characteristics inherited from parents can play a role in heart problems, but other causes come from the way people choose to live. Inactive lifestyles, obesity, the use of anabolic steroids (to increase muscle size) and drug use are examples of 'lifestyle' causes of heart disease.

- 1 What is meant by *obesity*? _____

Smoking is linked to many diseases, including heart disease. A heavy smoker who quits will have about half the risk of developing heart disease that a 'light' smoker who continues does. A person who never smokes is at the least risk of heart disease, provided other factors (such as heredity) do not play a role.

- 2 Suggest why people take up smoking or continue smoking even though they know it is harmful.
- _____



Other smoking-related conditions that can indirectly affect the heart include lung cancer, high blood pressure and, because of high levels of carbon monoxide (CO) circulating through the blood, cancers developing in major organs. A former smoker still has a high chance of developing cancers (though the risks are dramatically reduced by quitting) so it is most sensible not to begin smoking in the first place. Many public places are now smoke-free because 'second-hand' smoke (smoke exhaled by smokers) is highly toxic if inhaled. Constant exposure to second-hand smoke can lead to cancer developing in a non-smoker.

- 3 What does it mean to say something *indirectly* causes harm? _____
- 4 Describe one way to help warn people of the dangers of smoking. Outline the steps you would take, the main message and the audience your message is aimed at.
- _____
- _____
- 5 A person may never smoke, but still develop a smoking-related heart disease. Explain why this may happen.
- _____
- 6 Name four causes of heart disease, not related to smoking. _____
- _____

You may have heard the phrase 'hardening of the arteries.' The technical term for this is *atherosclerosis*. As we get older, fatty deposits can build up in the walls of our arteries and this reduces the ease of the flow of blood through blood vessels, causing the heart to work harder. If a clump of these deposits detach from an artery wall, they can cause a blockage which can result in a stroke. Cholesterol is a major contributor to this disease. Treatments include drugs, surgical removal of the affected parts of the arteries or mechanically 'opening up' the clogged arteries.

- 7 What do the following define?
- a** A harmful growth or tumour caused by abnormal and uncontrolled cell division. _____
- b** The build-up of fatty deposits on the linings of blood vessels, causing them to narrow and restrict blood flow.
- _____
- c** A condition where the body is unable to properly regulate the amount of glucose in the blood.
- _____
- 8 There are around 130 000 deaths each year in Australia. Cardiovascular problems (e.g. heart attacks) account for 38% of all deaths. Calculate how many people each year die from cardiovascular problems.
- _____
- 9 Why do some people need to check the level of cholesterol in their blood? _____
- _____



Arrhythmia

Any sort of irregular heartbeat can be classified as *arrhythmia* (*a* meaning 'lack', *rhythm* meaning 'pace'). The causes of this vary, but include the use of drugs, thyroid disease, disturbances to the body's ion balances and inflamed or diseased heart muscle tissue. Some of the effects of irregular heartbeat are breathlessness, 'cold sweats', palpitations (a 'pounding' heart), dizziness and varying degrees of chest pain. Arrhythmia is detected using a machine called an electrocardiogram, or EKG. (EKG is derived from the German term, *Elektrokardiogramm*.) This machine has detected irregularities in the heartbeats of patients who were unaware that there was a problem.

- 10** Why might it be important to know that a person has an irregular heartbeat?
-

Ischaemia (*iskhaimos*, meaning 'to stop blood') describes a condition where insufficient amounts of blood reach the heart. When normal blood flow is interrupted in this way, death or debilitation may result.

- 11** What is meant by *debilitation*?
-

Ischaemia can cause muscle fibres to beat 'out of tune'. This irregular beat is called *fibrillation* (*fibri* meaning 'fibres'), a dangerous version of a 'fluttering heart'. During fibrillation, the fibres of heart muscle quiver and 'short circuit', sending out electric charges at an unpredictable and irregular rate. Fibrillation is described as a sudden heart 'rush' followed by too-long a rest. Atrial fibrillation sometimes leads to the development of blood clots. Blood clots, or *thrombus*, are dangerous anywhere in the cardiovascular system, but the risk is greatest when they form in the major organs. This is because doctors have less time to isolate and deal with them before damage occurs. Blood clots in the heart can prevent the release of some, or all, of the blood from the atrium causing it to accumulate in the chambers.

- 12** Suggest why blood clots in a major organ might be dangerous.
-

- 13 Research** – Aspirin has been promoted by some medical practitioners as a means of preventing blood clotting and heart attacks. Find out more about the theory behind this.

Tachycardia & Bradycardia

When the heart beats too rapidly, the condition is called *tachycardia* (*tachy*, meaning 'swift'). Too slow a beat is *bradycardia* (*bradus*, meaning 'slow'). Common symptoms of tachycardia include dizziness and breaking out in sweats. The causes include drug use, alcohol and caffeine, smoking and an overactive thyroid (hyperthyroidism). Common symptoms of bradycardia include constant tiredness and feeling light-headed. The causes of tachycardia and bradycardia vary, and in many cases, the irregular heartbeat can be put right using medication.

- 14** Our hearts beat faster after exercise. Is this considered tachycardia? Explain.
-

Lasers can be used to treat tachycardia in cases where the unruly electrical charges travel the same, identifiable, route. The technique is called *radiofrequency ablation*. It uses lasers at a certain radiofrequency to burn the defective channels. The treated tissue is unable to transmit any more unruly electrical charges. Generally the operation is painless (at worst, uncomfortable) with a success rate of over 90%. However, this treatment is not suitable for everyone. When treatments fail, biomedical devices are an option.

- 15** What do the following define? Choose from: *arrhythmia*, *bradycardia*, *electrocardiogram*, *fibrillation*, *laser*, *thrombus*.

- a** A clot of blood formed within a blood vessel, that stays where it is formed.
- b** Failure of the cardiac pacemaker to maintain a regular heartbeat.
- c** Relatively slow heart action.
- d** Machine used to trace the behaviour of the electrical impulses in the heart.
- e** Acronym for light amplification by stimulated emission of radiation.
- f** When the muscle fibres of the heart contract in an irregular manner, resulting in the heartbeat and pulse being out of step.



For a person to be in good health, the heart must beat steadily and blood must flow continuously through it. At rest, the heart generally beats around 70 times a minute, governed by electrical stimuli generated by a region of the heart called the sinoatrial (SA) node. Sometimes these stimuli do not occur in the regular manner, and this can cause heart problems. Until about 40 years ago, major surgery was the only option for people with heart problems. The development of non-invasive and minimally invasive medical techniques has been of great benefit to patients.

- 1 Outline some advantages of non-invasive and minimally invasive medical techniques over earlier methods.

One of the most familiar biomedical devices is the *artificial cardiac pacemaker*. It consists of a corrosion-free lithium battery and a pulse generator that connects a wire to the heart. It works by sending an electrical charge from the site of the SA node.

The pacemaker came about quite by accident in 1958. While building a device intended to amplify and record the sounds made by hearts, American scientist Wilson Greatbatch invented a pacemaker. Modifications were made to the original invention, and the pacemaker became a viable proposition. The earliest pacemakers were only partially implanted. They had wires that emitted the electrical charge connected to an exterior device. Within ten years, totally implantable pacemakers had replaced these versions. In the past, the functioning of pacemakers was affected by the wearer standing too close to microwaves, airport metal detectors or other radio-type devices. These difficulties have been addressed, leading to the development of the superior models currently in use.

- 2 Suggest risks faced by individuals who used the earliest models of the pacemaker.

Modern pacemakers can detect when a person is overexerting, and the device increases its pace accordingly. The newest models can accumulate and store important details about the heart's activity which enables the physician to keep track of what has happened between visits.

There are two types of pacemakers – the single chamber, used for one heart chamber, and the double chamber, used for two. In general, pacemakers last 7–15 years, though manufacturers are constantly working to extend this time. While most people requiring them can use the single-chambered device, the more complex double-chambered version is not tolerated by all heart patients. Even a very old person can benefit from having a pacemaker implanted, provided they are healthy enough to survive the minimally invasive operation.

- 3 Figure 1 shows a simplified version of a typical modern pacemaker. Which component takes up the most space? Suggest why.

- 4 Comment on the properties of a material that could be used to make the case of a pacemaker.

- 5 The pacemaker is connected to the heart using insulated electrodes. Why must the electrodes be insulated?

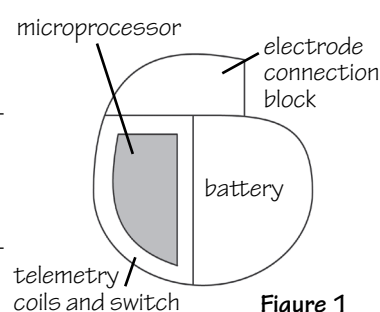


Figure 1

- 6 To what does the term *cardiac* refer? _____
- 7 Describe one other function of a modern pacemaker, apart from stimulating the heart. _____
- 8 How long does a typical pacemaker last? _____
- 9 Mark the following statements as true or false. Then correct the false statements so that they are true.
- a The earliest cardiac pacemakers were the size of a TV set. _____
- b Pacemakers give out exactly the same electrical signals, regardless of what the patient is doing. _____
- c The heart's 'natural' pacemaker is located in the sinoventricular node. _____
- 10 Which is the better treatment for heart disease – a heart transplant or a pacemaker? Explain your answer. _____



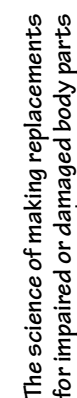
Concept maps are a useful way to summarise scientific (and other) areas of study. They show how various facts and concepts are linked. Making concept maps is a very efficient way to revise a topic. This exercise will help you to make your own concept maps for other themes.

On the next page there is a ‘skeleton’ concept map. Cut out the boxes at the bottom of this page and arrange them in the correct spaces on the skeleton concept map. Once you are certain you have the boxes in the correct spaces, glue them in place. Then answer the questions below.

- 1 Why is a concept map more useful than a list of ‘dot points’?

- 2 Add two more facts to the bionics concept map.
- 3 Choose one bionic device and create a concept map to show what you know about it. Now research 10 more facts about this device to add to your concept map.

Artificial heart valves	Biomaterials	Biomedical devices	Bionics
Cartilage replacement	Coat other materials	Heart, valve, hip joint	Immune reactions
Implants	Non-stick surface on pots and pans	Pins, plates and screws	Prosthetic cement
Prostheses	Teflon	Ultra High Molecular Weight Polyethylene	Pyrolytic carbon



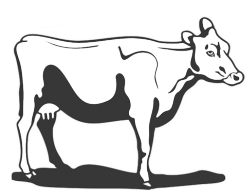


- 1 Write each term beside the correct definition: *alleles, genotype, heterozygous, homologous, homozygous, phenotype, dominant*.
- a The genetic make-up of an individual organism. _____
 - b Visible physical properties of an organism. _____
 - c Having two different alleles on homologous chromosomes for a characteristic. _____
 - d Different genes that can affect a particular trait. _____
 - e Having the same genes arranged in the same order. _____
 - f Having the same alleles on homologous chromosomes for a characteristic. _____
 - g The characteristic that shows up when there are two competing alleles. _____

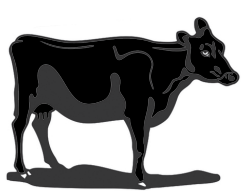
Usually when there are two alleles for a characteristic, one is dominant and ‘wins’ at being expressed, while the other is recessive and ‘loses’. This results in an organism showing one characteristic or the other. The recessive gene is still present in the organism, it is just not obvious to the naked eye.

Take the example of human parents, where one parent has blue eyes and the other has brown eyes. The offspring will usually have either two blue eyes or two brown eyes. It is very unlikely that an individual will have one blue eye and one brown eye, or eyes that are half brown and half blue – one characteristic is expressed.

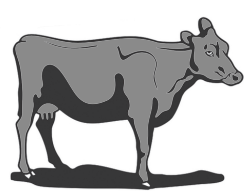
Sometimes single genes do not dominate. In *co-dominance*, both genes are expressed in the physical characteristics of the individual. There is not a blend of a characteristic, but a mixture – some of each gene is expressed.



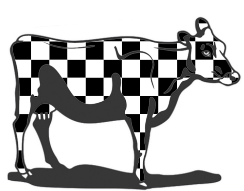
White cow



Red cow



Blended colour



Mixed colour

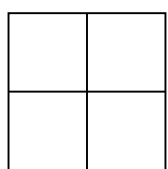
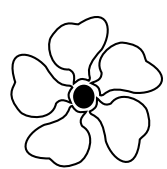
If a white cow and a red cow mate, there are various possibilities for the colour of offspring.

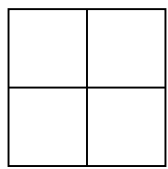
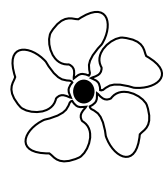
If conventional dominance occurs, offspring will be either white or red.

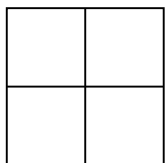
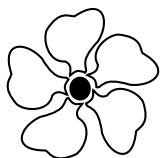
If incomplete dominance occurs, then individual hairs will be mid-way between white and red.

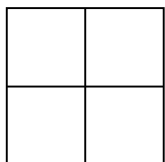
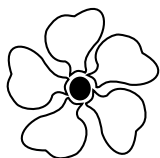
In co-dominance, some hairs will be white and some will be red. At a distance, the effects of co-dominance and incomplete dominance will seem similar. In cattle and horses, *roan* refers to a basic colour of white, mixed with other colours.

- 2 On the flower outlines below, show the possible phenotypes resulting from mating between red and white flowers, where incomplete dominance and co-dominance may occur. Complete a punnet square for each flower to show your work and include descriptions of what the colour combinations would be.









Environment and phenotypes

The phenotypes of an individual are its visible physical characteristics. Genes direct how an individual develops, and so direct the phenotype. The environment can encourage or restrict this development. If a gene is to be expressed, many conditions must be satisfied. In nature, environmental factors can affect physical development. These factors could include amount of nutrition, shelter and care from relatives.

- 3 Outline what is meant by a 'biotic' factor. Include an example. _____

- 4 What is meant by *visible*? _____
- 5 List two examples of phenotypic features. _____
- 6 Explain what is meant by *nutrition*. _____
- 7 How might nutrition affect development? _____
- 8 For each of the following examples, state what environmental factors would be necessary for the full potential of the genetic characteristic to be developed.
 - a Seeds of high-yielding strains of wheat. _____
 - b Seed potatoes that seem to be ideal for national marketing.

 - c Budgerigars that have parents that show signs of talking.

 - d A border collie that has parents who were champions at rounding up sheep.

 - e A young merino lamb that has a family with a history of very fine wool.

- 9 Discuss the idea 'nature vs nurture'. Include an example to clarify your discussion.

- 10 Circle the examples of phenotype that might not fully represent a person's genotype.
• eye colour • skin colour • rare blood disorder • height
- 11 List three other examples where a person's phenotype might not be fully expressed.

Figure 1 shows the type and percentage of blood groups in the Australian population.

- 1 List the groups in order of increasing size (i.e. least to most common).

The relative frequencies of the various blood types vary around the world. Karl Landsteiner (1930 Nobel Prize winner for Medicine or Physiology) classified blood groups in 1901, and discovered that they were inherited according to Mendelian principles.

When transfusing blood, it is very important to match the blood group of the donor and the recipient. If they are not compatible, the recipient can become ill or even die. This is because antibodies and antigens from different blood types can react with each other.

- 2 Use table 1 to explain why:

a Blood group O is often called the *universal donor*.

b Blood group AB is often called the universal recipient.

There are three major alleles that describe the inheritance of ABO blood groups. They can be represented as I^A , I^B and i . Table 2 shows the blood groups expressed by combinations of these alleles.

- 3 Referring to table 2, state which allele is dominant or if co-dominance occurs.

a I^A or i

b I^B or i

c I^A or I^B

- 4 Protective protein produced by the body as a response to an antigen – what does this define?

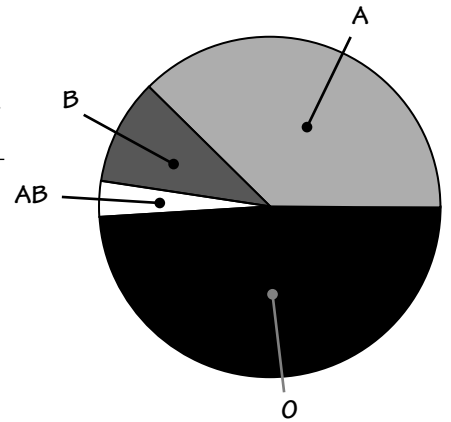
- 5 Find out the blood groups of the people in your class. In your workbooks, draw up a table to summarise your findings and convert the table into a pie chart. Comment on how (or if) your pie chart differs from figure 1.

- 6 Find out what *allogenic* means. What is an allogenic blood transfusion?

- 7 Name the four major blood groups.

Our blood groups are determined by tiny details on the surfaces of our blood cells.

The ABO system is the most important of the blood group systems. In addition to the ‘A’ and ‘B’ antigens designated by the ABO system, there is another series of antigens. These are known as rhesus (Rh) blood factors (named after the rhesus monkey, where this type of antigen was first discovered). The Rh group is the most complex of the blood groups. It is made up of several antigens. There are at least three separate genes close together on the same chromosome producing these antigens.



Donors	Recipients				
		A	B	AB	O
	A	✓	✗	✓	✗
	B	✗	✓	✓	✗
	AB	✗	✗	✓	✗
	O	✓	✓	✓	✓

Table 1

Blood group	Alleles
O	ii
A	$I^A I^A$ or $I^A i$
B	$I^B I^B$ or $I^B i$
AB	$I^A I^B$

Table 2



- 8

What is the name of the phenomena when there are more than two alleles coding for a particular phenotype?
- 9

A foreign substance that stimulates an immune response from the body, for example, by producing an antibody – what does this define?

Rh – The Rh-groups are the second most important of the blood groups. Medical practitioners classify people as being either Rh-positive (the Rh antigen is present) or Rh-negative (the Rh antigen is absent). Most of the population is Rh-positive. If Rh-positive blood is given to an Rh-negative recipient, no problems occur in the first donation. However, anti-Rh antibodies are formed, so if a second donation of Rh-positive blood is given, an adverse reaction (clumping) can occur. Similar problems can sometimes happen if an Rh-negative female has an Rh-positive child by an Rh-positive male.

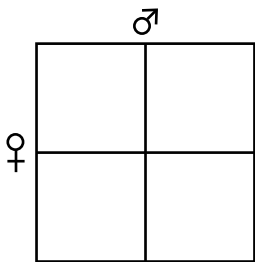
The allele for Rh-positive is dominant, and can be denoted as *Rh*. The Rh-negative allele is recessive, and can be shown as *rh*. Note that in this case, we use a two letter symbol for each allele. The *r* and the *h* do not stand for different things; they are part of the same single allele. Table 3 shows the phenotypes and genotypes involved in rhesus blood factors.

Phenotype	Genotype
Rh ⁺	RhRh or Rhrh
Rh ⁻	<i>rh</i> <i>rh</i>

Table 3

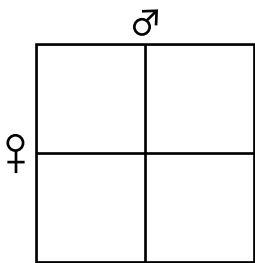
- 10

Using Punnett squares, show the possible genotypes and phenotypes of the offspring of the following:
a Father who is homozygous with respect to Rh-positive antigens and mother who is heterozygous with respect to Rh-positive antigens.



- b**

Father who is heterozygous with respect to Rh-positive antigens and mother who is Rh-negative.



- 11

What role does blood play in the body?
- 12

What might happen if a person gets a transfusion of the wrong type of the blood?
- 13

What do the symbols ♀ and ♂ mean?
- 14

Research – The International Society of Blood Transfusions recognises 29 major blood groups. Find out what these groups are.

Cells can be divided into two categories: *somatic* and *gametic*. Somatic cells are cells that make up body tissues, organs and all other parts of an organism, apart from the sex cells. Somatic comes from the Greek *soma* meaning ‘body’. Gametic cells are sex cells, such as sperm and ova. The word comes from the Greek *gamos* meaning ‘marriage’.

Cells that have two of each type of chromosome are said to be *diploid*. Somatic cells are diploid (Greek *diploos* meaning ‘twice’). Haploid cells have only one of each type of chromosome, half the number that are present in diploid cells. Gametes are haploid (Greek *haploos* meaning ‘single’).

1 Although gametes have only half the number of chromosomes as somatic cells, the zygote formed when an ovum is fertilised by a sperm will be diploid. Explain why.

Monohybrid crosses are crosses that involve only a single phenotype. But in reality, crosses involve more than one physical characteristic. *Mono* means ‘one’ and *hybrid* offspring ‘of one race’, so monohybrids are those offspring produced from one phenotype. Crosses that involve two characteristics are called *dihybrids*. *Di* means ‘two’ so dihybrids result from crossing two phenotypes.

2 Austrian botanist, who through his breeding experiments with peas laid out the science of genetics – who does this describe?

3 List an example of a monohybrid cross.

4 List an example of a dihybrid cross.

Among other things, Gregor Mendel studied these crosses in garden peas: round vs wrinkled seeds, green vs yellow seeds, green vs yellow pods, red vs white flowers.

He found that dihybrid crosses produced offspring that had phenotypes different to the parents. For example, the genes for seed colour and the genes for seed appearance were on the chromosomes of different homologous pairs, that is, were independent.

5 The Punnett square below outlines the results of crossing two pea plants. The characteristics studied in these seeds are round (R) vs wrinkled (r) and green (g) vs yellow (G). Complete the genotypes and phenotypes.

		Male: RrGg (round, yellow)			
		RG	Rg	rG	rg
Female: RrGg (round, yellow)	RG	RRGG round yellow	RRgG		
	Rg				
	rG				
	rg				rrgg

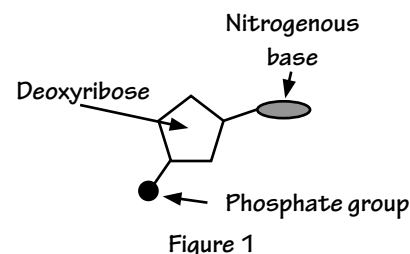
Hint: To help you remember which cells carry half the genetic material, and which carry the full amount, think: **d**iploid **d**ouble, **h**aploid **h**alf.



Chromosomes are the structures that are responsible for passing on characteristics from one generation to the next. Chromosomes are always present in the nuclei of cells, but are only apparent when cell division is occurring. They need to be stained to be visible.

- 1 A sequence of nucleotides on a chromosome, forming the functional units of inheritance for a particular characteristic – what does this define?

A chromosome is mainly composed of a DNA (deoxyribonucleic acid) molecule. DNA is a very large molecule, containing many thousands of atoms. This in turn is composed of individual units called nucleotides. A nucleotide is a repeating sub-unit of nucleic acids. There are four major nucleotides in DNA, repeated many times. Figure 1 shows the general structure of a nucleotide.



- 2 Name the three constituent parts of a nucleotide.

There are four major nucleotides in a DNA molecule. All contain the phosphate and deoxyribose (sugar) groups, but have different nitrogenous bases. The nucleotides are held together by bonds between the deoxyribose and phosphate groups. The different nitrogenous bases are adenine, guanine, cytosine and thymine.

- 3 Figure 2 shows a portion of a single chain of DNA. Identify what is signified by:

A _____
G _____
T _____
C _____

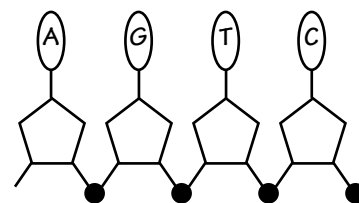


Figure 2

The sequence of the nucleotides determines the specificity of the DNA. Figure 3 shows how the nucleotides in the two chains of the double helix are held together by (weak) hydrogen bonds.

- 4 Identify the hydrogen bonds in figure 3.
5 What is the pattern of linkages between adenine, guanine, cytosine and thymine?

- 6 What is the type of bonding between the nucleotide groups?

- 7 Notice that A and T 'pair up', as do C and G. A student used the mnemonic 'Aunt Tracey Cooks Grapes' to help her remember the pairing. Make up your own mnemonic for this.

- 8 In any DNA molecule the amounts of adenine and thymine are always equal and the amounts of cytosine and guanine are also equal. Explain why.

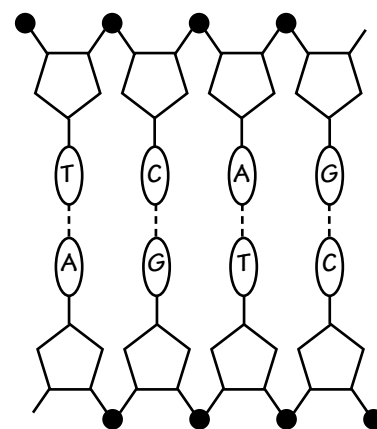


Figure 3



The word *recombinant* comes from ‘to recombine’ or ‘to combine again’. Recombinant DNA is made by splitting DNA molecules and then ‘splicing’ together specific DNA fragments from two or more DNA fragments. In other words, recombining fragments to produce a new combination.

DNA can recombine naturally, as it does during the joining of two haploid cells. However, recombination can also be achieved artificially.

This technique has given biologists a greater understanding of the make-up of genes and how they function. A simplified sequence of the process is shown in figure 1.

Note that in biology, a vector can be an organism that carries and transmits disease. In genetics, it is a carrier piece of DNA.

The donor DNA and the vector (usually a plasmid from bacteria) are split using microbial enzymes (endonucleases). The highly specific nature of these enzymes causes the cut ends of both sets of DNA pieces to match.

- 1
- If the sticky end from the donor fragment has a nucleotide sequence of ATAT, what will be the sequence of the sticky end of the open plasmid?
-

The ‘foreign’ gene sample is spliced into the open plasmid, which is then re-joined using DNA ligase.

- 2
- What type of compound is a *ligase*?
-

This modified plasmid is put into a bacterial cell, such as *E. Coli*. The recombinant DNA is reproduced exactly inside the bacterium cell. This cell then divides to form genetically identical copies (clones) through its normal replication process. The altered plasmids are then removed from the bacterium cell. Using the original restriction enzyme, these plasmids are again split and the genes copied from original donor DNA are obtained in large quantities.

The cloned gene is then analysed to find its amino acid sequence. It can be broken down by hydrolysis and the fragments identified. The position of this gene on the chromosome can be found by making a radioactive probe. This probe is made up of the complementary base pairs of the recombinant DNA and will latch onto the equivalent site on the chromosome. Autoradiography (technique similar to X-ray photography) shows the position of the binding site.

- 3
- Why is the probe radioactive?
-
- 4
- What do these describe?
- a

A circular DNA molecule that replicates independently of the bacterial chromosome
- b

Making genetically identical cells or organisms asexually
- c

Piece of synthetic DNA that has been made radioactive
- d

Cell with only one full set of chromosomes
- e

Biological catalyst
- f

Phosphate units, ribose sugar units linked with nitrogen bases

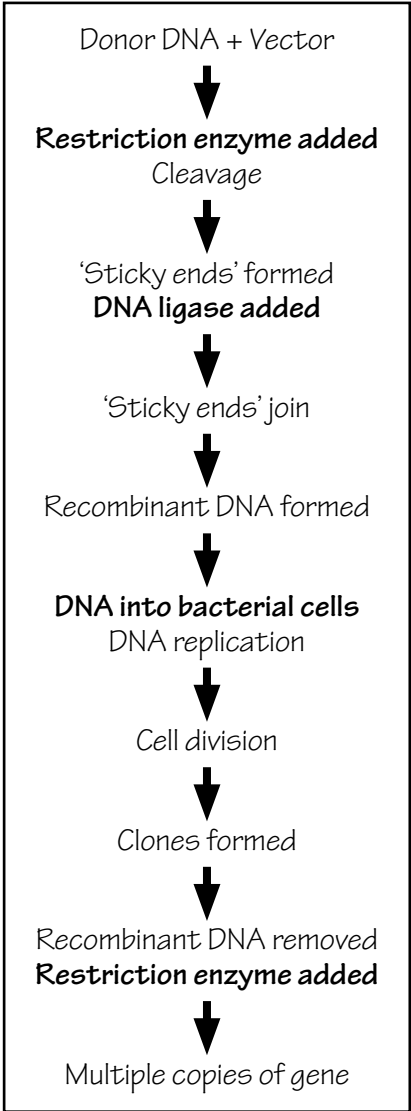


Figure 1



- 1

Define *cloning*.
- 1

Write each term beside the correct definition: *somatic cell, differentiated, clone, genome, stem cell, culture, germ cell, in vitro*.

a

The growth of cells, tissues or whole embryos in an artificial medium.

b

Complete genetic make-up of a plant or animal.

c

Cells that grow into sperm or eggs – ‘sex cells’.

d

An exact replica of a cell, tissue, organ, plant or animal.

e

‘Body’ cells. Cells not involved in reproduction (e.g. not sex cells).

f

To have developed into a specific type of cell or tissue.

g

Cells with the potential to develop into any kind of cell type.

h

‘In glass’ using glass dish, test tube or other artificial environment.

- 2

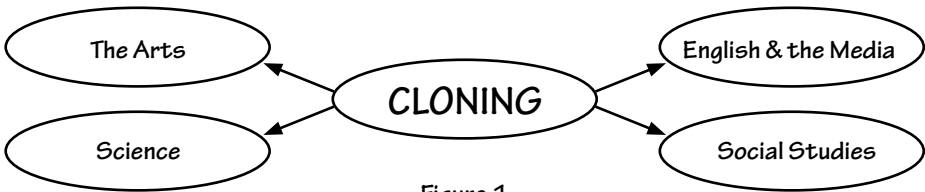
List five things you know about cloning and five things you want to find out about it.

Things I already know about cloning

Things I want to find out about cloning

- 3

Examine the fields shown in figure 1. Consider how each one might present the theme of cloning to the public. Then complete the tasks below in your workbook.



The Arts Imagine you are the curator of a museum in which cloning is the theme of a special exhibit. How might different artists present their ideas about cloning? Sketch a statue, painting or another artform through which artists might express their feelings about cloning.	English and the Media Write a news headline and the first paragraph of a journalism article about cloning (no need to write the whole story). Alternatively, write the title and cover blurb for a fiction or non-fiction book about cloning. (It could be science fiction!)
Science Use dot points to show how scientists might present the issue of cloning to the public, including arguments for and against.	Social Studies Consider the perspective on cloning of a person who specialises in social studies, or of someone in the general public. Sketch an image or write words they might use to describe cloning.



1 – Classifying diseases

- Teacher to check, e.g. lung cancer from passive smoking, skin cancer from sun exposure
- Few people intend to, many are the result of lifestyle habits e.g. drinking alcohol.
- e.g. Tobacco mosaic virus, mould, rust
- Teacher to check. **a** Lifestyle/mental illness **b** Degenerative **c** Dietary, mental illness **d** Environmental
- Congenital/ environmental/ lifestyle
- Contagious **g** Contagious
- Lifestyle/ dietary/ congenital
- Congenital **j** Congenital
- Congenital/ lifestyle/degenerative
- Contagious/lifestyle **m** Contagious
- Congenital/environment **o** Lifestyle/ dietary **p** Degenerative **q** Mental illness **r** Mental illness **s** Contagious
- Occupational/congenital **u** Usually mental illness, also lifestyle as there is evidence of marijuana inducing this.
- Dietary/lifestyle
- Teacher to check
- Some are not self-inflicted in one group and are in another; e.g. heart disease can be congenital or self-inflicted as a result of obesity, same for diabetes
- They impair overall health and wellbeing
- Teacher to check

2 – What is disease?

- a** Epidemic **b** Contagious **c** Hypochondriac **d** Ailment **e** Pandemic **f** Epidemiology **g** Pathogen
- Odds are only one parent carries a rare condition
- Teacher to check; e.g. PKU, albinism, cystic fibrosis, haemophilia
- 3 250 000
- Poor health because of reduced nutrients
- A parasite
- Teacher to check **8** Teacher to check
- Jaws lock, paralysis, can't eat
- Water and air
- Yes, 'many lives have been saved'
- Gonorrhea, not everyone shows symptoms, those who do show burning, discharge from genitals. Student may pick another, e.g. genital herpes, blisters, syphilis
- Teacher to check
- Obese is grossly/excessively overweight

3 – Disease in the community

- Kills the conditions in which it thrives
- To reduce the risk of contracting airborne or contact disease organisms
- Discharge transferred by direct contact and flies
- Teacher to check
- Chickens and pigs are often found near each other, farmers feed dead chickens to

pigs. Other animals might also be if they are kept near chickens

- Airborne or by being in contact with droppings
- Organ will have donor's proteins on it, not 'self', rejection
- How many people (out of a given number infected) die of that specific cause
- Very rapid and severe
- The strain of bird 'flu they contract will be resistant to the vaccine
- Mutation
- Crowded conditions, almost certainly come into contact with virus
- Discuss economic effects, fear among society, etc.
- Breathing, lungs
- Aches, chills, fever, cough, sneezing
- Atypical pneumonia, originate from China, corona virus
- a** False **b** True **c** False **d** True **e** True **f** False **g** False **h** True **i** True
- Teacher to check

4 – Disease – vocabulary

- a** Symptom **b** Chronic **c** Disease **d** Contagious **e** Sign **f** Infect **g** Systemic **h** Primary **i** Colonise **j** Secondary **k** Local **l** Acute
- In order: Disease, symptom, sign, infection, contagious, colonised, local, systemic, primary, secondary, acute, chronic
- a** True **b** True **c** True **d** False **e** True **f** False **g** False **h** True **i** True **j** False **k** True
- e.g. Mosquito-borne diseases more common in tropical and subtropical areas because that is where they thrive
- Different initial health levels, care available
- Teacher to check, e.g. rabies

5 – Antibiotics and disease

- 35% of the people who contract diphtheria will die of the disease
- a** Easily 'caught' **b** Refers to only one thing **c** Happened before
- Initially high numbers of deaths, then the numbers drop very low and then slowly rise.
- Between 1935 and 1945, as the number of deaths decreased
- Between 1955 and 1965, as the number of deaths fell dramatically
- Bacteria, virus would not have been affected by antibiotic
- Number of deaths is rising, perhaps current antibiotic is losing effectiveness
- 3000, 6000+ approx.; extrapolate figures from data
- Inflammation of heart/nervous system
- Teacher to check

6 – Disease – graphing

- Teacher to check
- Effects of the disease
- Days 5 to 14 **4** First four days
- Around 700, lowest amount when symptoms are present
- Not fair to ask anyone else, wanted to feel effects himself. Possible risk to health

7 – Invasive and non-invasive procedures

- Teacher to check. Extreme distress, disturbance to the body's natural systems, etc.
- Teacher to check, e.g. removal of appendix
- Responsible for the production of certain hormones
- Family and patient are worried financial being the cost of paying medical staff, electricity, machinery costs, etc.
- e.g. Laparoscopy is now minimally invasive but in the past, tying tubes involved a major operation, some heart operations
- To prevent pain, reduce patient movement
- Teacher to check; e.g. increased chance of diagnosis, reduced number of unnecessary surgeries.
- Any form of radiation that forms ions, e.g. pulls electrons from atoms to produce electrically charged particles
- e.g. Alpha or beta particles, gamma rays, x-rays
- e.g. Can cause tissue damage, lead to cancerous growths
- e.g. To check for abnormalities, see if baby is developing properly, age/sex of foetus
- e.g. Damaged tissue from severe strain or break
- $C_5H_9N_3$ **14** e.g. Allergies, asthma
- a** Blood vessels **b** Childbirth **c** Eyes
- Check health of the heart, check for damage, etc.
- Something that conducts electricity. In this case can be from heart activity
- Improved diagnosis, hence treatment
- Not dense enough
- For gallstones
- Doesn't use potentially dangerous ionising radiation
- Can't show details inside the body
- 2D – length and breadth; 3D has an extra dimension, length, breadth and depth.
- Teacher to check details agree with text
- Cut
- A painful, chronic condition, caused by growth of endometrial cells outside the uterus, extreme cramping, etc.
- Local removes pain from an area, general means patient is unconscious
- Lead from ovary to uterus
- Teacher to check



Selected answers (cont'd)



30 Advantages: a, c, d, f

31 Include any signs of brain activity, other damage to body, and wishes of relatives

32 Teacher to check

8 – Medical procedures quiz

1 **a** True **b** True **c** False **d** True
e True **f** False **g** True

2 **a** A hand-held probe, used to direct the path of ultrasound sound waves

b Magnetic resonance imaging, magnetic force allows us to identify the properties of atoms, thus can produce an image of the tissue being scanned **c** Invasive involves more trauma, often larger incisions, etc. minimally invasive involves using natural body openings or keyhole surgery, reduced trauma, etc.

d Cardiac ultrasound. Uses echoes to create an image of the heart **e** High frequency sound waves cause leucocytes to release histamines. Histamines increase diameter and permeability of blood vessels to increase blood flow to area. Histamines attract monocytes to clean up the site and activate fibroblasts which secrete collagen to heal tissue fibres

3 **a** Obstetric **b** Sonography
c Vascular **d** Ophthalmic

4 Teacher to check

5 Electrodes conduct electricity. They can be used to detect brain or heart activity, not usually considered invasive

6 X-ray, exposes patient to ionising radiation

7 Not usually but it is useful in detecting gallstones

8 **a** Tomography **b** Atoms
c Bioscopy **d** Heat **e** Density
f Invasive

9 – Ethics

1 Might be biased or seen to be biased

2 e.g. Is it necessary to use animals? Can results be reached another, more humane way? What is the minimum number of animals required that would still give a valid outcome? What considerations will be made to keep the subjects as comfortable as possible?

3 Include person's wishes, quality of life, life is sacred

4 Illegal and unethical, perhaps should have checked

5 TGA, Therapeutic Goods Administration

6 For: helps science, might provide a cure for 'flu. Against: disrespectful, could release diseases

10 – Organ transplants

1 e.g. it was alien, not the hand he 'knew', etc. Frightened him to have a dead man's hand

2 Teacher to check, e.g. no one knows how the frozen tissue will function, etc.

3 Teacher to check

4 Infection, power cuts, machinery failure, patient's movement restricted

5 Cultural, religious reasons

6 e.g. Can't really enjoy life, Dracula was neither dead nor alive; Schroeder would be breathing, etc., but not enjoying life as he wanted

7 e.g. Being housebound, worrying about having artificial components

8 To be discussed. Was it worth it to be alive longer with loved ones etc vs dying earlier? Helping advance science, etc.

9 Teacher to check

10 Weighed about 1.5 kg, battery operated, rechargeable by painlessly sending 20 watts of power through the skin, portable power pack enables heart to pump for several hours before needing to be recharged

11 – DNA on the footy field

1 Fitness, size, coordination, mental attitude, speed, muscle tone

2 How well fed, exercised and cared for. Education, mixing with other children, medical care

3 e.g. Twin one may lack confidence in what could be good abilities; poor diet could affect his/her health and fitness, etc. Or twin one might be more determined to succeed than twin two

4 Teacher to check. Should give a reasoned argument

5 In order: outlook, enthusiasm, personality

6 DNA won't necessarily expose these, too complex

7 e.g. Football players have a certain mindset – it's the game they love, not athletics per se

8 Could easily be injured on the field or outside the game, for example in a car crash or by taking illegal recreational drugs

9 May affect other life choices the individual makes, could be depressed or try harder

10 Mental covers things like determination vs physical, is pure strength without considering emotions. Mental strength can push a person to work harder

11 e.g. Potential height, speed, strength. Physical things based on a person having good diet and sufficient exercise.

12 e.g. Some might be labelled inferior when they could be excellent players. Lost opportunities, unfair favouritism

13 Teacher to check 14 Teacher to check

15 Teacher to check, but is outside accepted values

16 Couples being fertile, staying together, rights of child, female children

17 Both are looking at potential, traditional is more concrete, DNA more 'in the future'

12 – Medical technology – bionics

1 Similar to humans

2 e.g. Giving them superhuman abilities

3 e.g. Darth Vader, Dr Who's Daleks, 6 Million Dollar Man, Robocop, the Borg

4 Teacher to check

5 e.g. Hearing aid, pacemaker, spectacles, dentures, crutches, walking stick, cochlear implant, hip replacement

6 Teacher to check

7 **a** Implant **b** Biomaterial

c Prosthesis **d** Cyborg

8 Seek and kill invaders/disease

9 So they don't react with the body's biochemical reactions, don't rust, etc.

10 Metal combines with oxygen and forms a new compound

11 Immune system will initiate an attack response and the body might reject the device

12 Biomaterials are the materials used to make biomedical devices

13 A mix of metals chosen to combine the most desirable properties of each metal present

14 **a** Biocompatibility **b** Low friction
c Strong **d** Durable **e** Unreactive
f Lightweight

15 e.g. Biology and physiology to understand how a healthy heart works. Developing the materials used for the device requires an appreciation of the body's immune system, to ensure the body does not reject it. A knowledge of chemistry is essential to create the materials. Physics and Engineering are necessary to design the mechanics of the device and all branches use mathematical modelling

16 Industry involved in design and development of aeronautics (aeroplanes) and astronautics (space travel)

17 e.g. Alloys and polymers

18 Cell, tissue, organ, organ system, body

19 It quickly detects invaders

13 – Biotechnology – A brief history

1 Damaged/weakened

2 To hold bones perfectly still, which helps a broken limb to mend

3 e.g. acupuncture, massage, leeches

4 Covering chips, discoloration, etc.

5 False teeth

6 e.g. Replacement joints, artificial limbs, replacement heart valves, pacemakers

7 Cosmetic, looks better than a closed or damaged eye

8 Vision is more complex than hearing

9 **a** Cosmetic **b** Bionic **c** Artefact

14 – Biotechnology – terms & materials

1 Can't get lost, contains lots of information, more expensive, more technology, can't immediately be seen
2 e.g. Computers, calculators, DVD players, iPods, anything 'electronic'



- 3 Could cause damage to eyes
4 C: *Mono-* meaning 'one', *chroma* meaning 'colour', consisting of one colour
5 Yes, it gets the engine started
6 Teacher to check, should start a device
7 **a** Polymer chain **b** Monomer
c Chemical bond
8 When pulled they stretch, when force is released they return to original size
9 Chemically inert, flexible
10 **a** Teacher to check **b** Teacher to check
11 Connective tissue
12 Teacher to check
13 Wool clothing and keeping sheep warm, silk clothing and parachutes
14 Sheep 15 Silkworm
16 **a** Durable **b** Versatile **c** Polymer
d (Blood) clot **e** Compatible
f Pyrolysis
17 For appearance
18 Silicon 19 Oxygen
20 Carbon 21 Not as good
22 Needs surgery
23 e.g. Device can become loose and cause injury
24 If cement used in dentistry needs replacing it can be done quite harmlessly and with patient awake whereas for implants, it's much trickier
25 Sets firmly, unreactive
26 e.g. Transparency, strength, low density
27 Teacher to check
28 Teacher to check
29 Protect brain tissue from external injury, blows, etc.
30 Component snaps, damage to bone, pain to person, new operation needed
31 Dentures complete replacement, can be removed, crown partial replacement, permanently fixed
32 Cosmetic

15 – Prosthetic devices

- 1 Limb, addition, attachment 2 e.g. Peg could be contaminated 3 e.g. Self esteem 4 No. A help, not a replacement
5 Haemorrhage, internal bleeding
6 Teacher to check. Mention cost, comfort, appearances 7 Get worse, deteriorate 8 These are all indications of circulatory problems 9 e.g. Risk of heavy bleeding from injuries
10 Arteries are bigger, carry larger volumes of blood 11 Teacher to check
12 **a** True **b** False **c** False **d** True
e True **f** True **g** False **h** True
i False **j** True **k** True
13 Engineering solutions for damaged bodies **a** Multidisciplinary **b** Tissue
c Corrosion **d** Alloy **e** Cosmetic
f Implanted **g** Teflon **h** Polymers
i Sutures **j** Egyptians **k** Thrombus

16 – Cardiovascular disorders

- 1 Extreme excess body weight in the form of fat 2 e.g. Peer pressure, advertising, addiction 3 e.g. It doesn't harm you directly but causes something to happen that does 4 Teacher to check
5 Passive/second-hand smoking
6 Characteristics inherited from parents, inactive lifestyles, obesity, the use of anabolic steroids and drug use
7 **a** Cancer **b** Atherosclerosis, coronary heart disease **c** Diabetes
8 49 400 9 Danger of clogged arteries
10 Could be the indication of more serious problems to come with the heart
11 Weakness 12 Block blood flow/oxygen 13 Aspirin has an anti-inflammatory effect, inhibits platelets from collecting to form clots 14 No, this is a normal reaction. Tachycardia is an unhealthy condition with an underlying medical cause 15 **a** Thrombus
b Arrhythmia **c** Bradycardia
d Electrocardiogram **e** Laser
f Fibrillation

17 – The artificial pacemaker

- 1 Less trauma to patient, less chance of infection 2 Affected by other electronics
3 Battery, provides energy for a long time
4 Would need to be inert/unreactive
5 So that electrical signals are only received by the heart 6 The heart
7 Collect data 8 7 to 15 years
9 **a** True **b** False **c** False, sinoatrial
10 Consider availability, timeframe, 'need'

18 – Concept map – bionics

- 1 Bionics 2 Biomaterials 3 Immune reactions 4 Biomedical devices
5 Prostheses 6 Implants
7–8 Prosthetic cement; pins, plates and screws 9 Pyrolytic carbon 10 Coat other materials 11 Ultra high molecular weight polyethylene 12 Cartilage replacement 13 Teflon 14 Artificial heart valves 15 Non-stick surface on pots and pans 16 Heart, valve, hip joint

19 – Genotypes and phenotypes

- 1 **a** Genotype **b** Phenotype
c Heterozygous **d** Alleles
e Homologous **f** Homozygous
g Dominant
2 Teacher to check. Will be: white, red, blended (pink) and mixed (red and white). Punnett squares should show an acceptable alternative to the conventional conventions of dominant (R/W) and recessive (r/w) and a reasonable depiction of offspring
3 Anything to do with life, e.g. characteristics of parents
4 Obvious, that can be seen
5 e.g. Hair colour, eye colour
6 Eating and utilising food
7 Too much, get fat, not enough too thin, wrong sort, dietary disorders
8 **a** Enough water and sunlight, freedom

- from weeds **b** Right type of soil, enough moisture **c** Owners that talk to them **d** Training with sheep **e** Right type of food, environment
9 Teacher to check, but should include relevant examples
10 Eye and skin colour and height
11 e.g. Weight, earlobes, tongue rolling

20 – Blood groups

- 1 AB, B, A, O 2 **a** People from all blood groups can accept a blood donation from a person with blood group O
b People from AB blood group can accept a blood donation from a person any blood group 3 **a** I^A is dominant
b I^B is dominant **c** Co-dominance
4 Antibody 5 Teacher to check
6 Genetically different, allogenic transfusions are those that involve giving mismatched blood 7 A, B, AB, O
8 Homozygous 9 Antigen
10 **a** Should be consistent with table 3; i.e. father RhRh and mother Rhrh
b Should be consistent with table, i.e. father RhRh and mother rhRh
11 Transports food, waste, oxygen, heat
12 The person will become ill, and possibly die. 13 Female, male

21 – Diploid and haploid

- 1 Two gametes are involved 2 Gregor Mendel 3 e.g. White and black fur
4 e.g. White and black fur plus blue and brown eyes 5 Top row: RRGg, round yellow; RRGg, round yellow; RrGG, round yellow; RrGg, round yellow.

Second row: RRGg, round yellow; RRgg, round green; RrgG, round yellow; Rrgg, round green.

Third row: RrGG, round yellow; RrGg, round yellow; rrGG, wrinkled yellow; rrGg, wrinkled yellow.

Fourth row: RrGg, round yellow; Rrgg, round green; rrGg, wrinkled yellow; rrgg, wrinkled green.

22 – Chromosomes and Genes

- 1 Gene 2 Phosphate group, nitrogenous base and deoxyribose
3 Adenine, Guanine, Thymine, Cytosine
4 Indicated by dashed lines on the diagram 5 A/T and G/C 6 Hydrogen bonding 7 Teacher to check
8 Because of A/T and G/C pairing

23 – Recombinant DNA

- 1 TATA 2 Enzyme 3 So it can be traced/located 4 **a** Plasmid
b Cloning **c** Probe **d** Haploid
e Enzyme **f** Nucleotide

24 – Cloning – points of view

- 1 Cloning is the production of genetically identical individuals by asexual means
2 **a** Culture **b** Genome **c** Germ cell
d Clone **e** Somatic cell
f Differentiated **g** Stem cell **h** In vitro

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